

```
//CSE  
110;
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UCD and Design Process Part 2

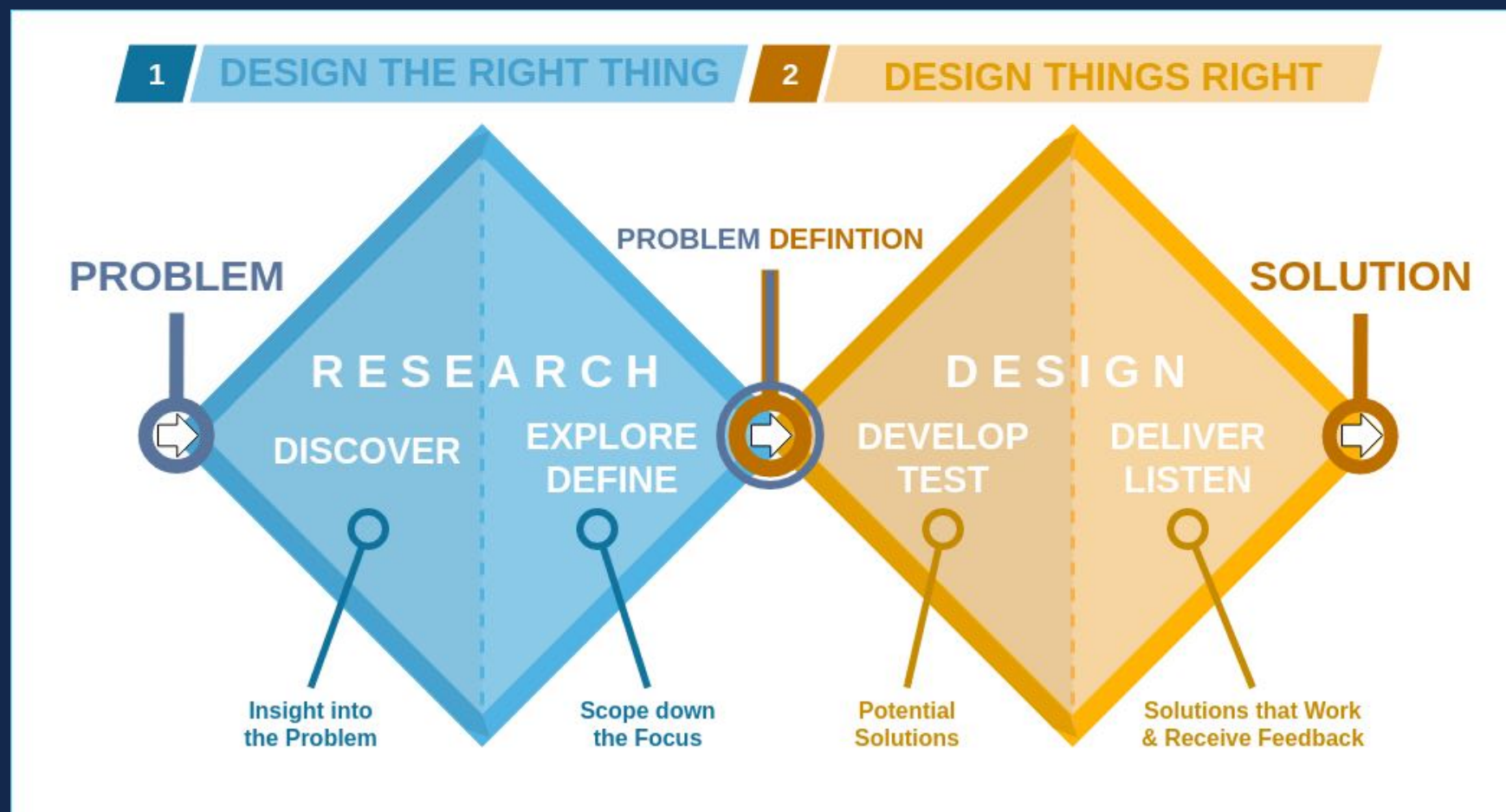
with Thomas Powell

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We Need a Process

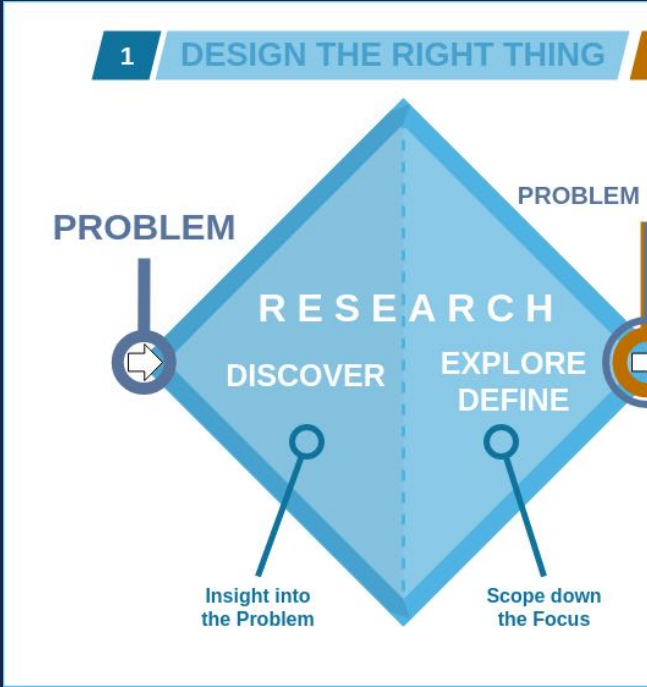
- The course will use an Agile model to conduct our business but what exactly are we building?
- Sure we have at this point a prompt, but we are likely not really ready to start writing code and setting features.

Let's Use the Double Diamond Process



Where are we now?

Let's Use the Double Diamond Process



First we go broad

Then we narrow down

Lots of Artifacts and Efforts

Landscape Research

Core Principles Defined

Technology Exploration

Persona Development

Journey Mapping

User Stories

What We Will Aim to Produce Alignment Diagrams

Rough User and Problem Exploration Plan

- Problem Definition - Prompt Explored and Refined, First Principles, Existing Solution Exploration
- Broad User Exploration - Gains, Pains, and Jobs to be Done
- Specific User Exploration - Personas (hopefully driven by data and actual users)
- Users Experience - Customer Journey Map, User Story Map
- User Activities - User Stories, Use Cases

Eventually the Problem is Defined

- We have a sense of who we are building for, what their problems are and roughly what they might need.
- Next we define the activities of our users
 - Users Experience - Customer Journey Map, User Story Map
 - User Activities - User Stories, Use Cases
- Finally we will now be ready to scope and design. We will soon see the use of the *Shape Up* principles to help us not go overboard with our design as we start planning our execution.

Fight Understanding Attrition with Alignment Artifacts

- Without artifacts capturing your ideas attrition of alignment is guaranteed as memory fades, participants depart, and situations change.
- An artifact we dub an Alignment Diagram is something concrete that can bring a team back to a shared view.
- An alignment diagram can help new members develop and add to the shared understanding
- This artifact can outlive a team shared mental understanding and is superior to just an understanding unsaid between devs or devs and stakeholders

Principles of Alignment Artifacts

- **Holism** - cover the whole thing
- **Multiplicity** - show all the multitudes of thoughts, feelings and actions
- **Interaction** - show the “touch points”
- **Visualization** - make sure it is visually pleasing
- **Self Evident** - make it obvious
- **Relevance** - be relevant to what needs to be understood
- **Validity** - be backed up by data where possible, not just opinion

The ABCDEs of A Good Alignment Diagram

- **Attractive** - Is appropriately visually attractive
- **Big** - Shows the big picture from product out to the world and back
- **Clear** - Is easy to comprehend
- **Diverse** - Shows different dimensions and types of information
- **Enough** - shows what is necessary to build alignment for everyone



Alignment documents are foundational.
Good solutions grow on top of them!

Example Alignment Diagram Types

Table 1. Different ways to diagram aspects of value-centered design

DIAGRAM TYPE	STORY	INTERACTION	INDIVIDUAL	ORGANIZATION
Customer Journey Map	Chronological	Touchpoints	Actions, thoughts, feelings, pain points, etc.	Roles and departments involved in creating an experience
Experience Map	Chronological	Touchpoints	Actions, thoughts, feelings, pain points	Physical and social artifacts in a system; opportunities
Service Blueprints	Chronological	Line of interaction	Actions, physical evidence	Backstage actors and processes
Mental Model Diagrams	Hierarchical	Center line	Tasks, feelings, philosophies	Support—products and services available
Spatial Maps	Spatial	Midpoint with arrows	Actions, needs, information flow	Data systems, departments

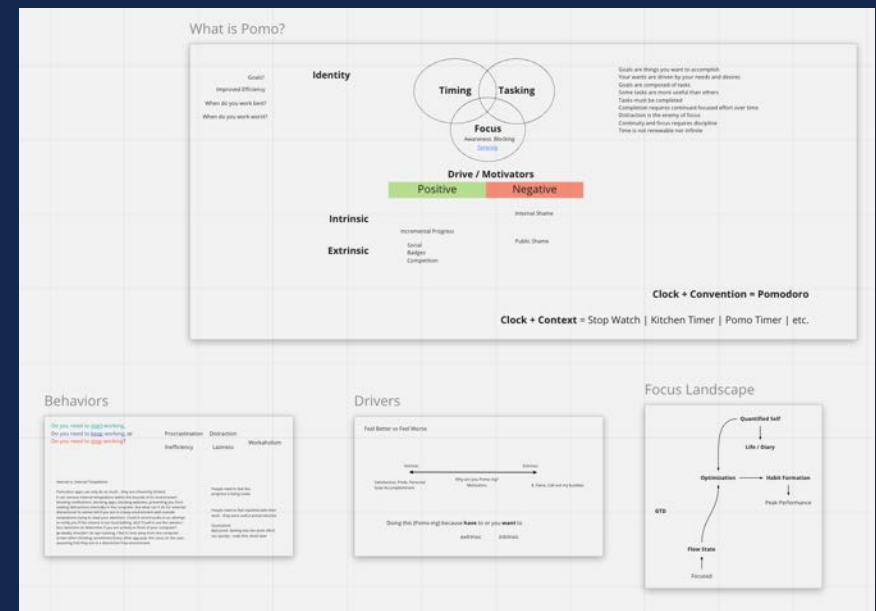
<https://www.oreilly.com/content/introducing-alignment-diagrams/>

Aligning Understanding

Broadly all this effort should be helping us define a shared understanding of the **WHO** we are serving and exactly **WHAT** problem we are solving for them and **WHEN** and **WHERE** they might engage with that problem. Then and only then might we start consider **HOW** we might accomplish our solution. If we don't answer these questions **WHY** we are doing it is more based upon instinct rather than upon careful thought.

Step One: Do a Decent Think About First Principles and Final Outcomes

- Remember the Musk - Bezos Battle?
- Reminder: Press Release Backwards vs First Principles Forwards?
- No. Do both!
- Define a time box (set time length) to do this effort. Eventually you will condense some of this into your overall design document or even pitch.



A **small** snippet of some of the thinking we did internally about the Pomo Timer

Step 2: Think Deeply about Users

- Now define the users you will serve and the goals they have, before enumerating the features you'll implement so those users can achieve those goals.
- Implementing before any or with very little user consideration is dangerous as you likely may build something they don't want or dislike.
- It is also a bit selfish as it focuses on **your need to make** first and foremost.
 - Define a time box (set time length) to do this effort
 - Many artifacts will likely be produced

Why You Must Do User Driven Development

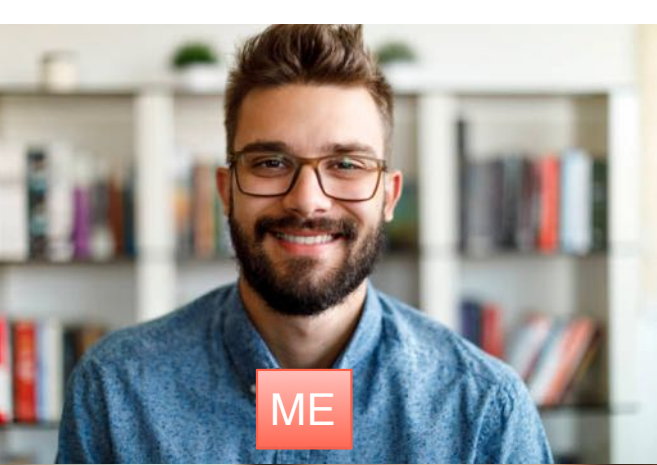
- For software where user choice is an option, **user acceptance should be considered the largest risk**
“You can do everything right and construct the software properly on time and budget and ultimately fail if the user’s do not accept it.”
- An outside-in approach, front-end first, or user driven development (UDD) approach should be employed in interface and market driven situations.
 - In theory all process models can support a UDD approach, as long as users are addressed early enough (specification) and often (validation and evolution)
- Be careful a UDD idea in the form of evolutionary prototyping, JAD, etc. can really be quite frustrating and impossible to budget for other than a time / resource cost point of view.

Let's Ask Chad to Help Us Break Feature First Thinking

The following slides show a visual demonstration of how we could deeply personify a study of users and needs if we wanted to. In this case I used a constant stock photo person to model their life and problems that some software might solve.



Hey there everybody my name is Chad Coder. I am originally from Stockphotostan. I currently live in Venice California and am a dev at [GettyImages.com](https://www.gettyimages.com)



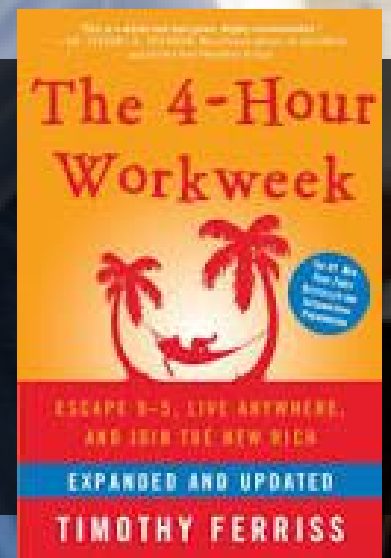
ME

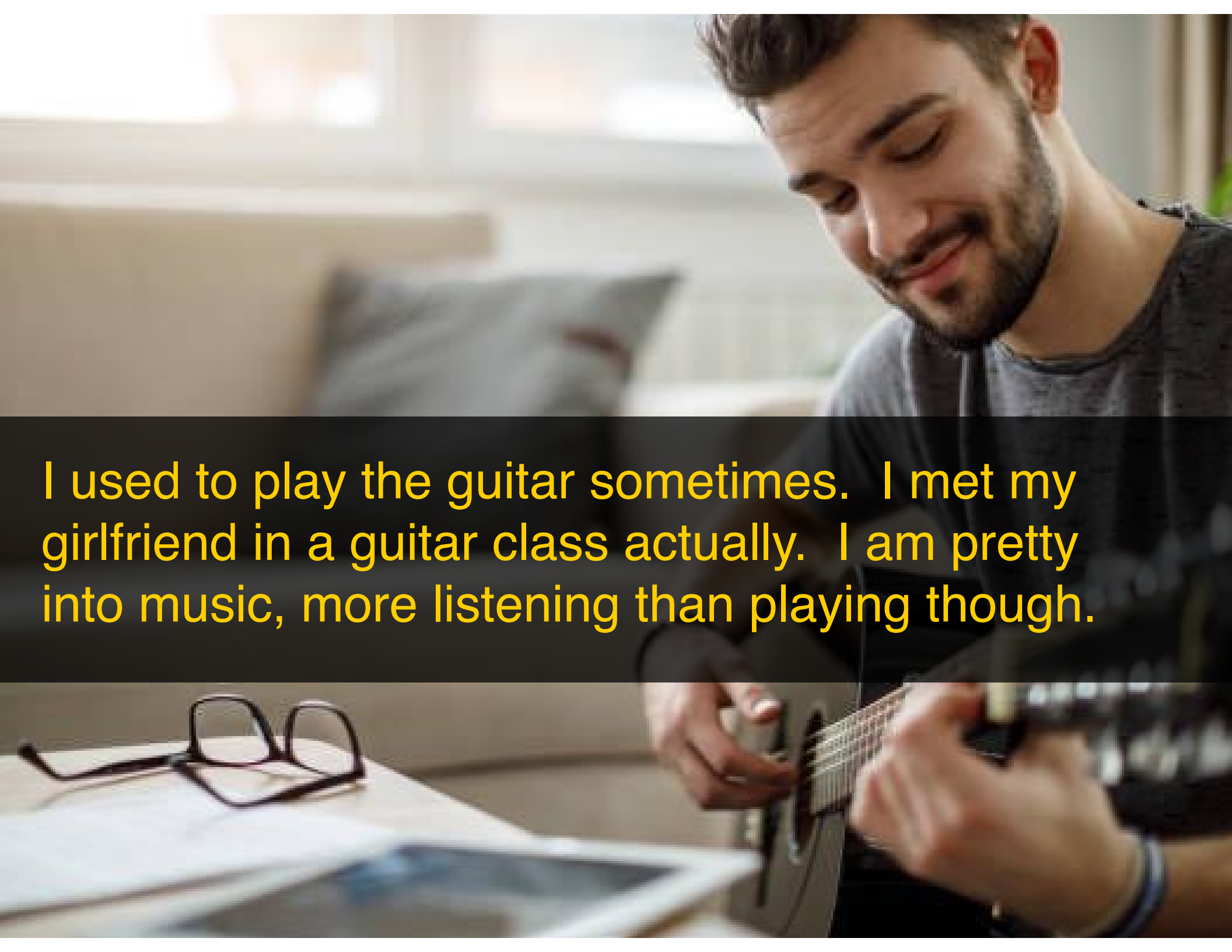


Famous Cousin

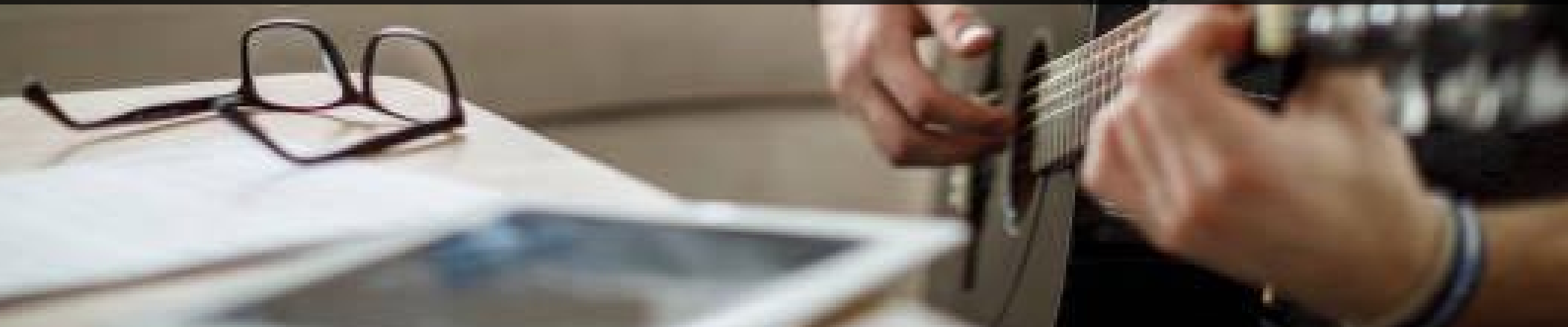
You probably know my cousin...he's kinda of
(in)famous

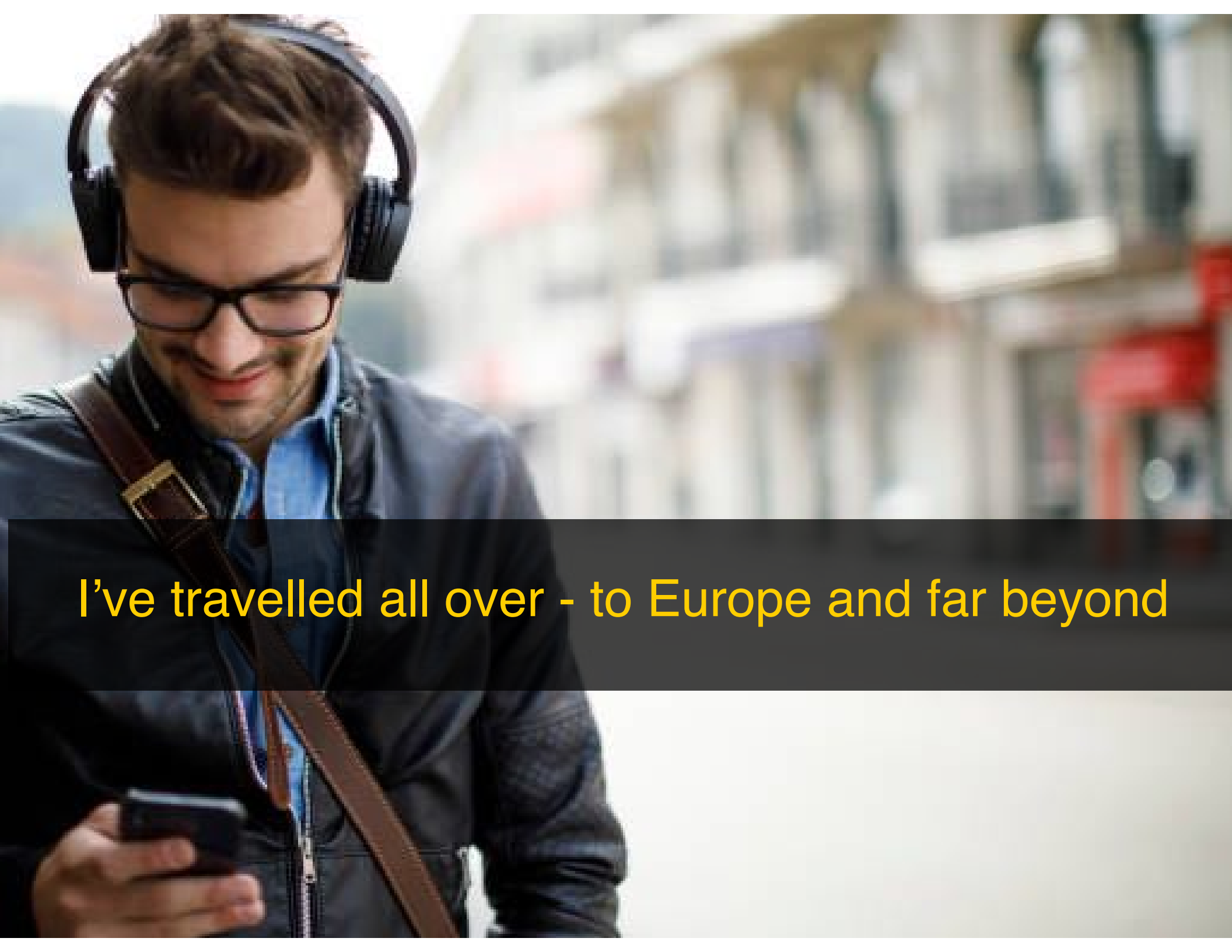
Pre-COVID I rocked some stubble and was doing that whole digital nomad thing. Tim Ferris is my spirit animal.



A man with a beard and short dark hair is playing an acoustic guitar. He is looking down at the instrument with a slight smile. He is wearing a dark grey t-shirt. The background is a blurred living room with a couch and a window. A semi-transparent black box with yellow text is overlaid on the middle of the image.

I used to play the guitar sometimes. I met my girlfriend in a guitar class actually. I am pretty into music, more listening than playing though.

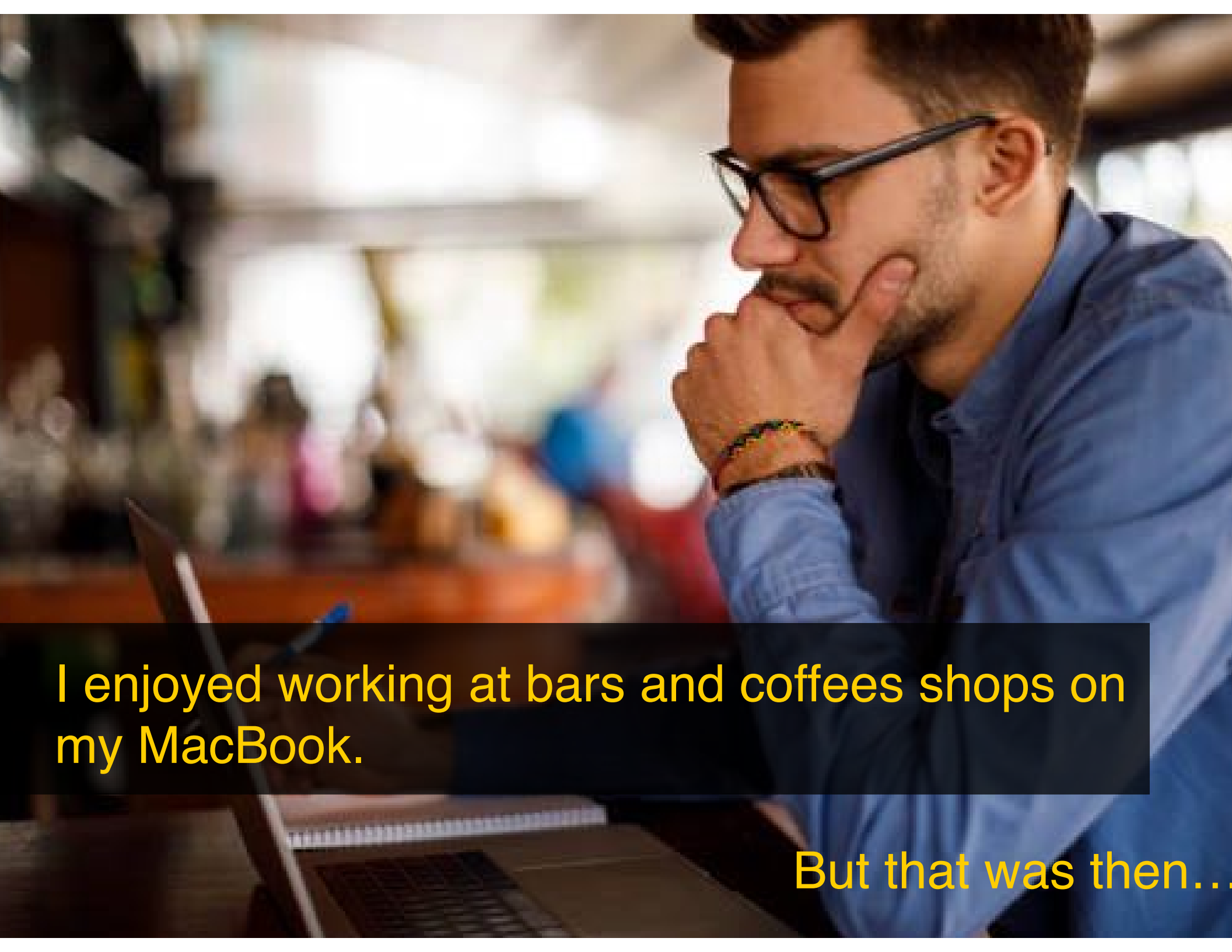




I've travelled all over - to Europe and far beyond



I remember I used to go outside all the time...
I even wore a hat sometimes!



I enjoyed working at bars and coffees shops on my MacBook.

But that was then...



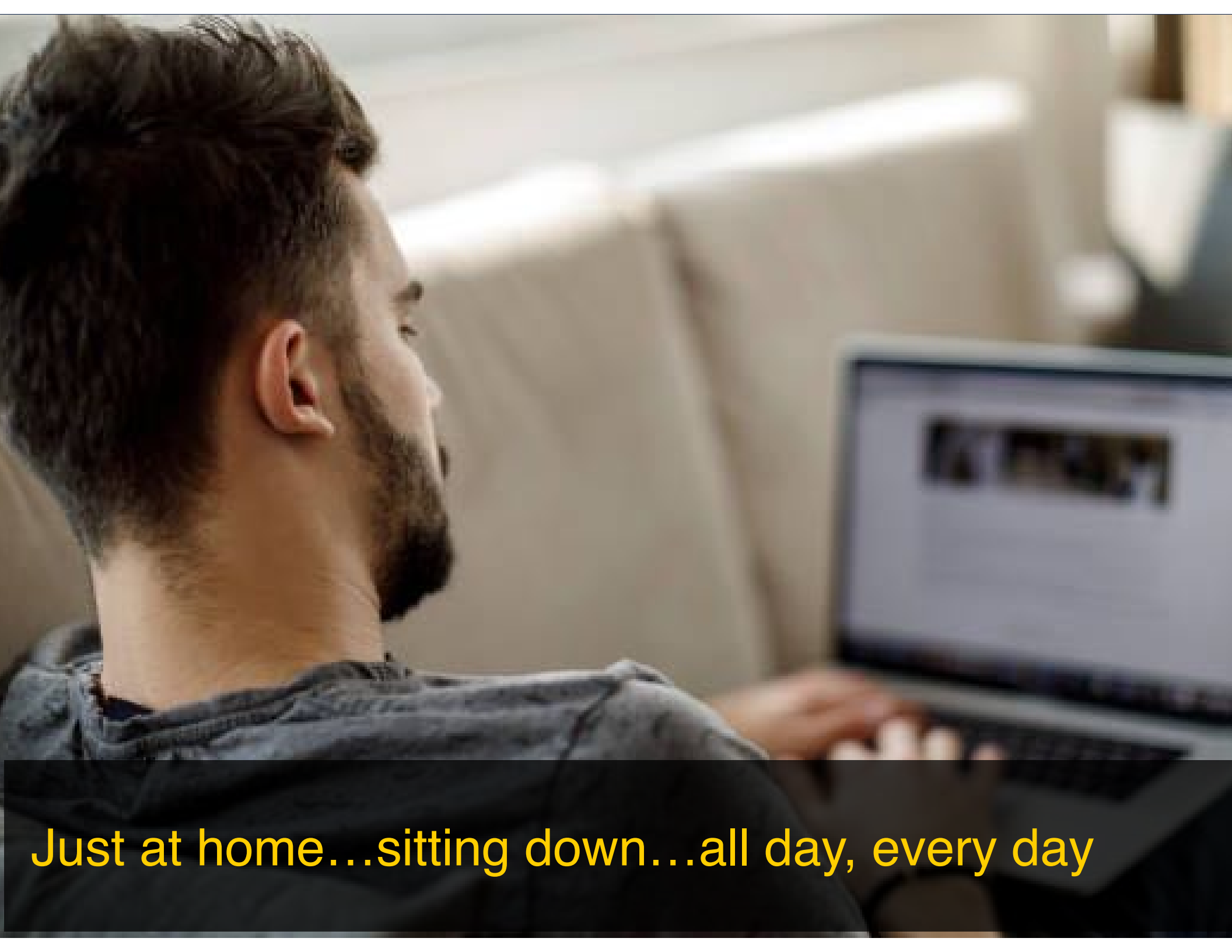
So the Professor Powell guy says you might be able to help me... you see these days I got a problem...

Not a health problem though, I always wear my mask and do my best to deal with our COVID world.



My problem started because I can't go places anymore...I'm stuck at home just like you.

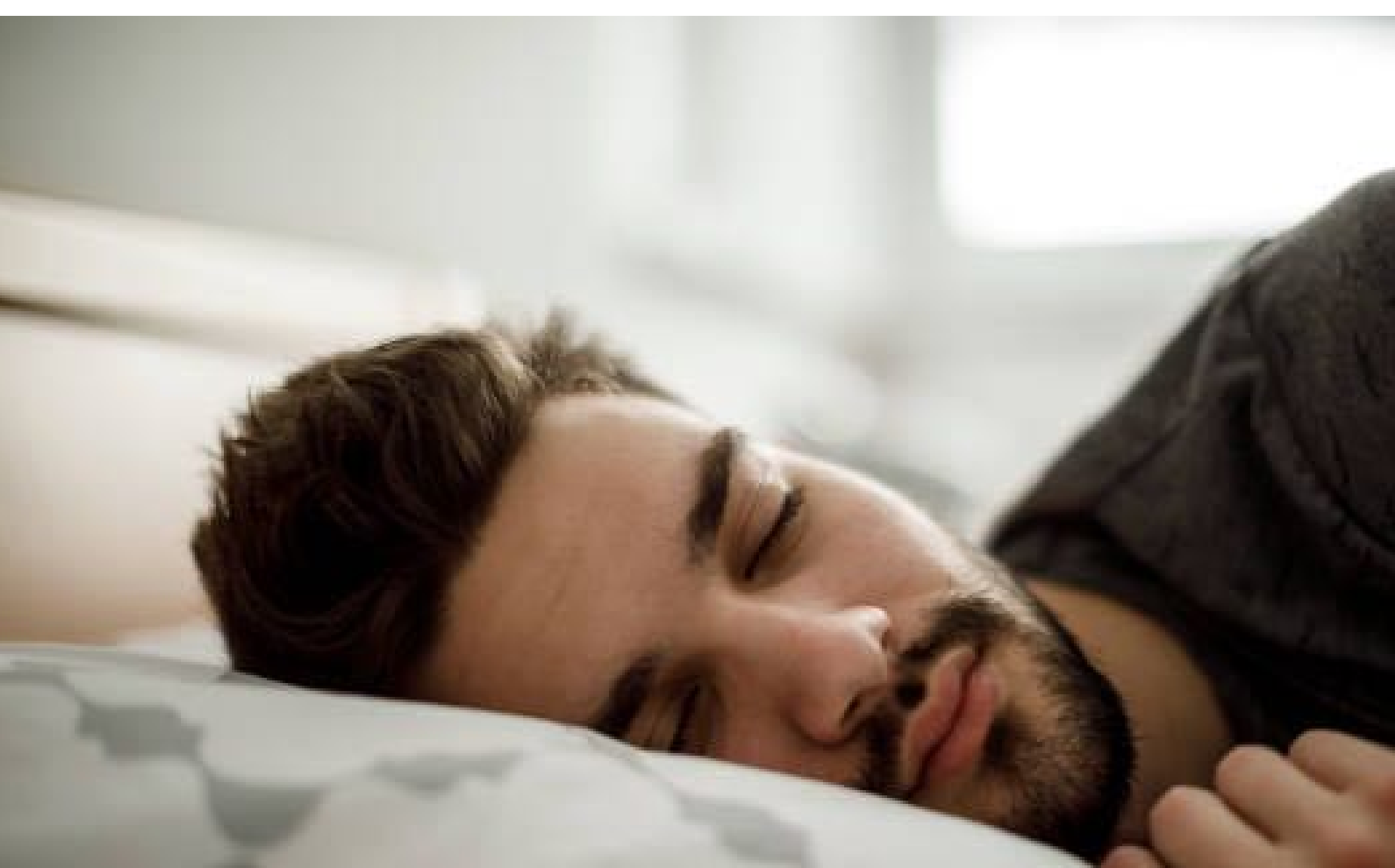




Just at home...sitting down...all day, every day



My day? Sure I'll tell you about my day, how I work and some other things I do.



My day starts when I wake up.

First action, the morning scroll begins.



I guess on a positive point I am like a semi-pro
Doom Scroller now after all the practice.



After a good scroll, I have a cup of coffee with my best pal Reddit.



Eventually, it's time to work!

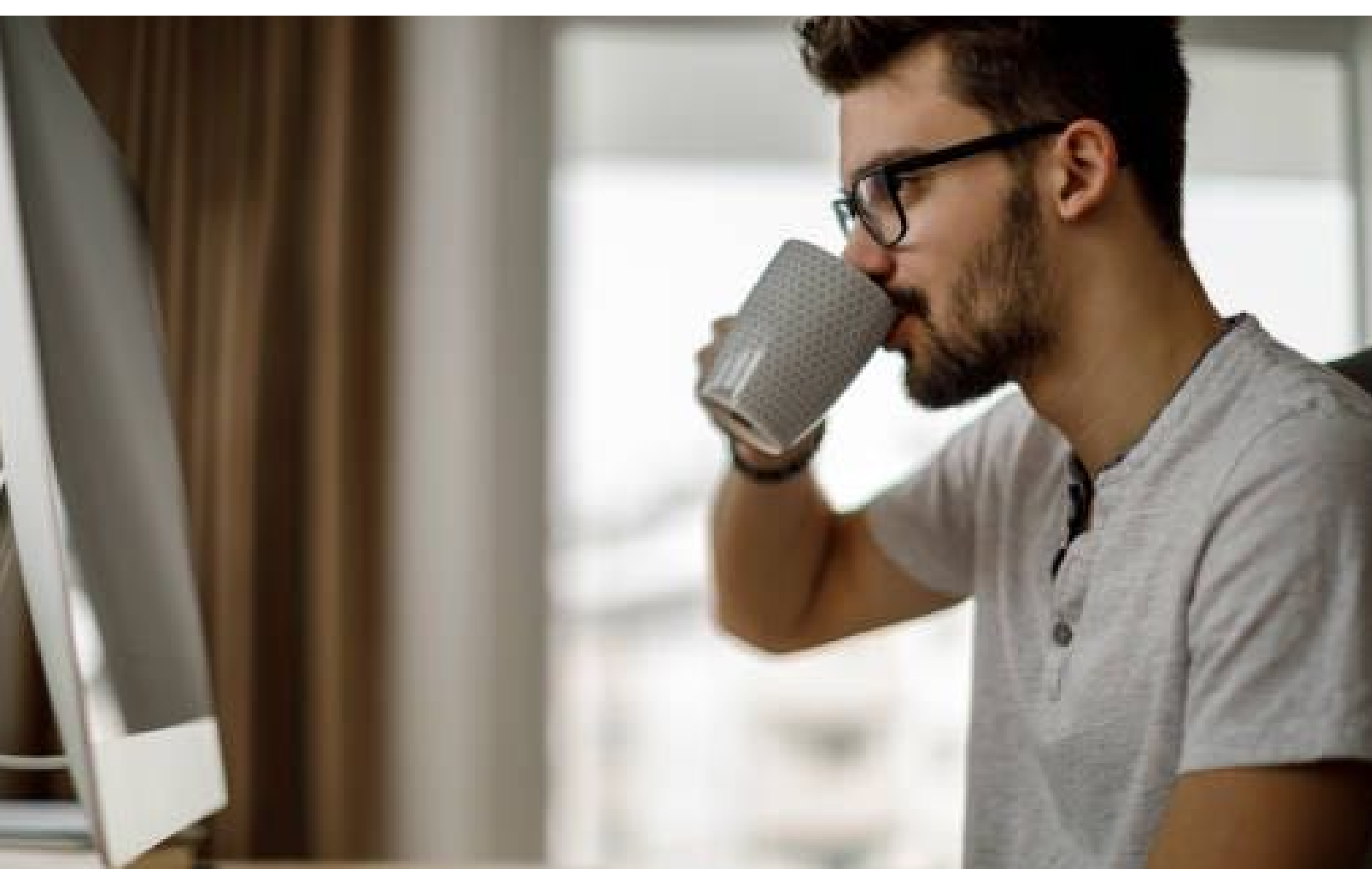


Until the group chat starts popping, then I gotta send some solid reaction GIFs.





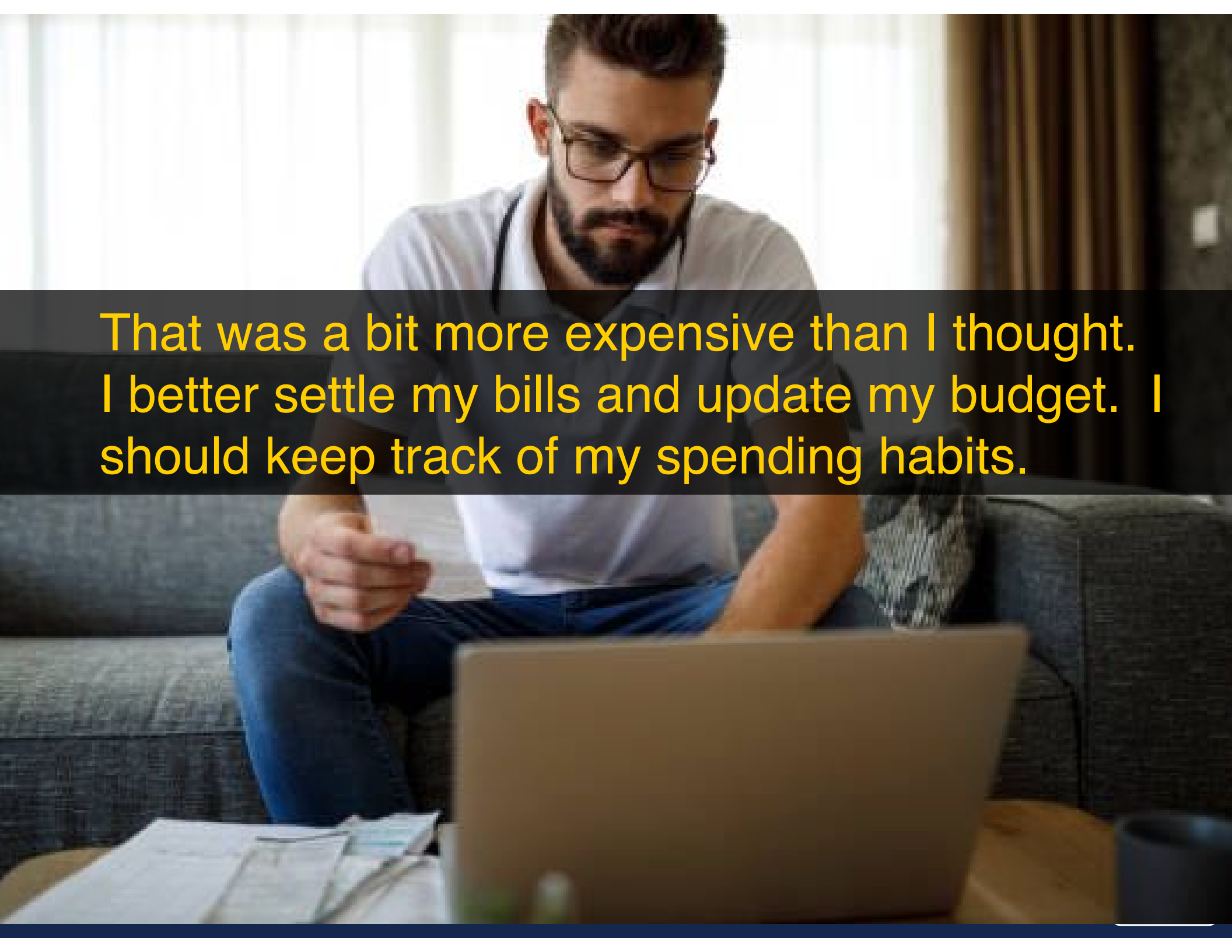
Wait now the boss is calling. Old fashioned phone call, but no camera so that's a plus.



Ok with that settled, time to get back to work.

A man with dark hair, a beard, and black-rimmed glasses is looking down at a smartphone in his hands. He is wearing a blue button-down shirt under a dark grey blazer. The background is softly blurred, showing what appears to be a window with light coming through. A semi-transparent dark banner with yellow text is overlaid across the middle of the image.

Oh I saw that hoodie in Instagram early. A little retail therapy is like a water break. Just a few minutes to get that done.

A man with a beard and glasses is sitting on a dark grey couch, looking down at a laptop screen. He is wearing a white t-shirt and blue jeans. His hands are clasped together in his lap. In the foreground, there are some papers and a small potted plant on a wooden surface. The background shows a window with light-colored curtains and a dark curtain on the right side.

That was a bit more expensive than I thought.
I better settle my bills and update my budget. I
should keep track of my spending habits.



Cranberry Juice skateboard guy... I love that one! What did Matt Stonie just post...60second Filet o' Fish Challenge!?!

Lunch Time! Ok, I device and eat, so what?



I lift and learn to keep the body and mind in shape at once. I probably should record my efforts to get a better sense of my progress.



Well since I twofered with lift and learn, I deserve another break. Clash Royale time!





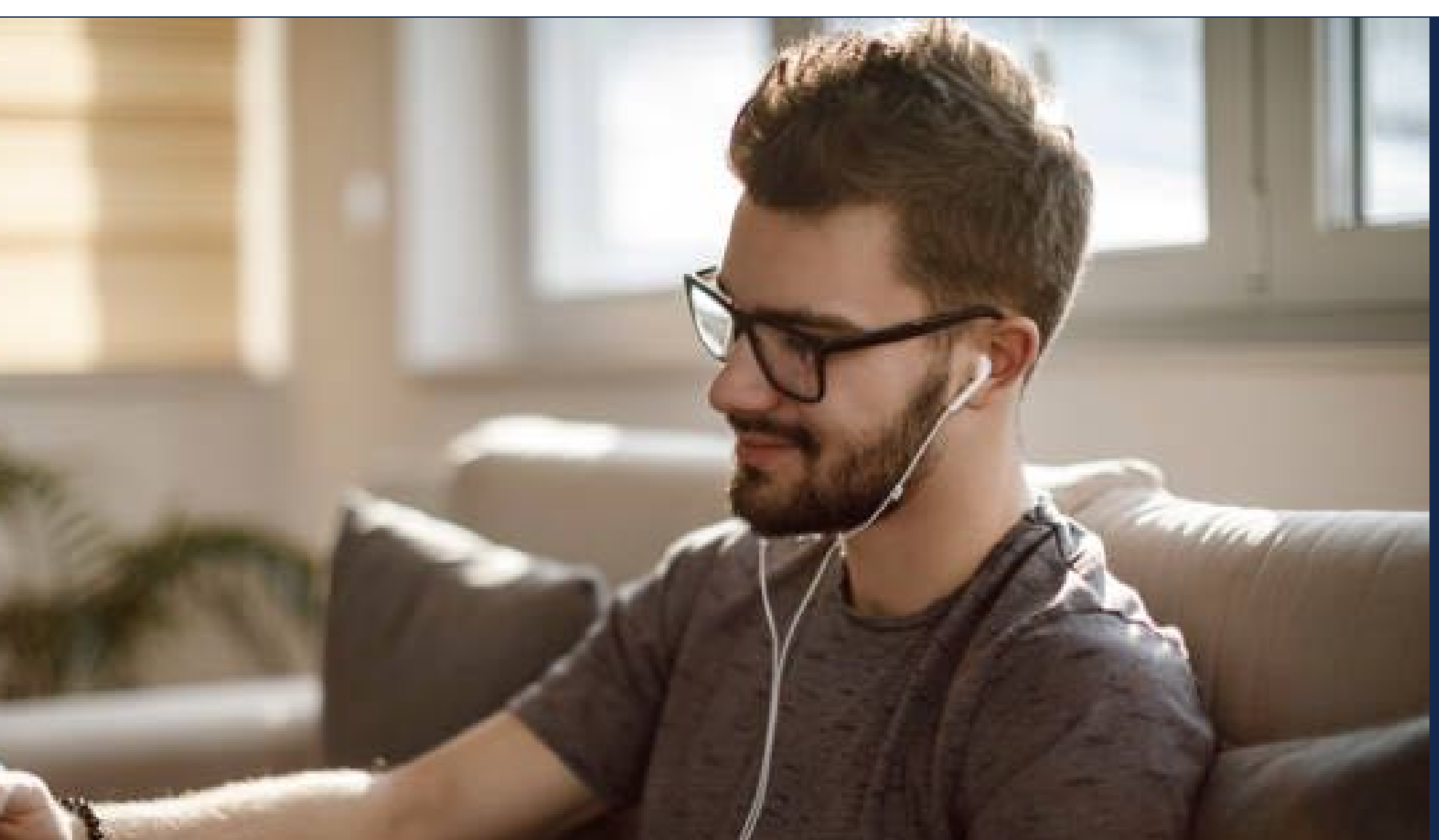
There is a lot more to me. I have hobbies, I like to read, play guitar, and oh I collect things... especially headphones.



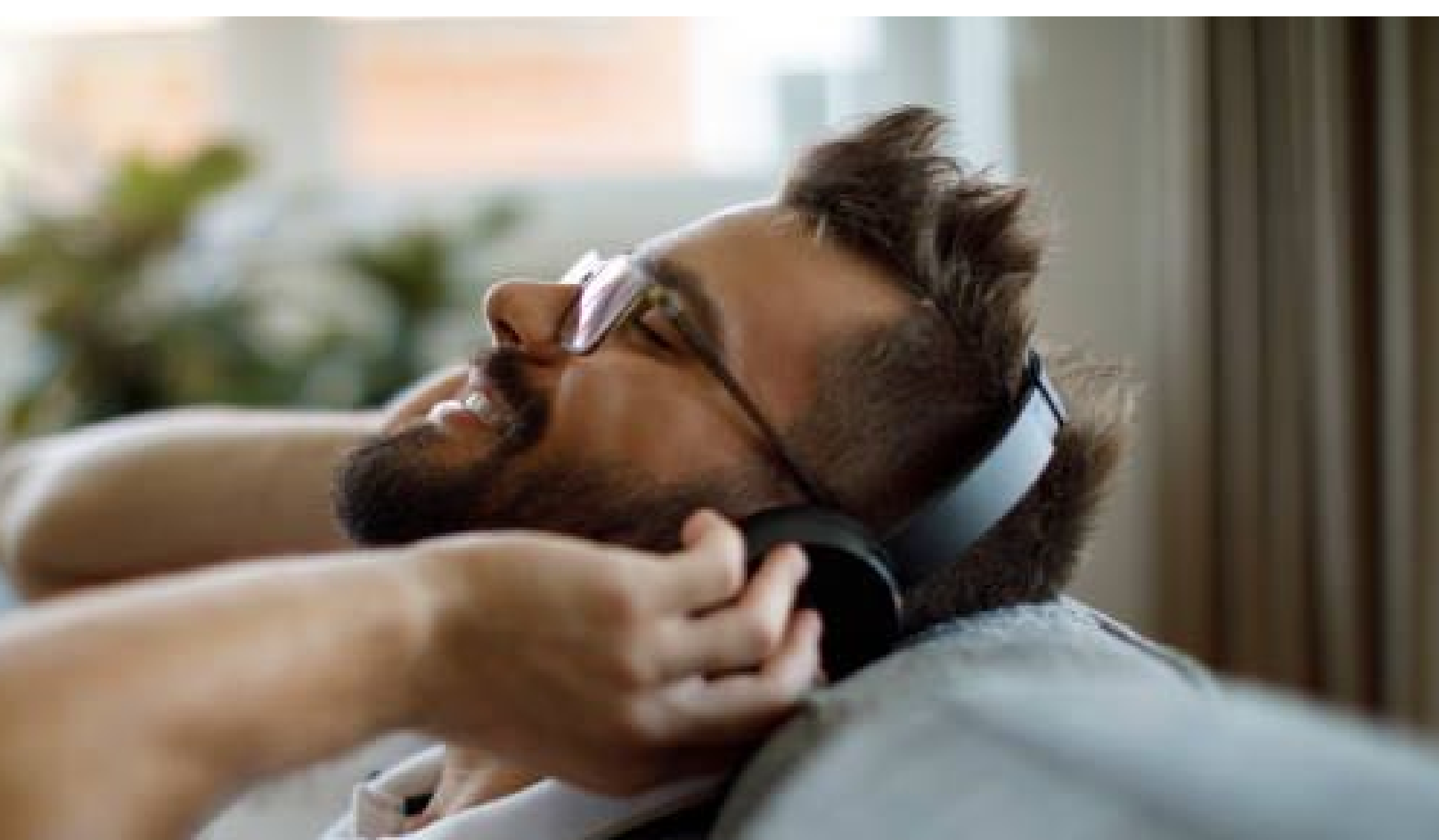
As if you haven't noticed yet I really like wearing headphones and I have lots of them.



On Zoom calls I usually rock the AirPods 2 buds



I know wired buds feel are retro, but I use them too especially when listening to audio books. I like the feel.



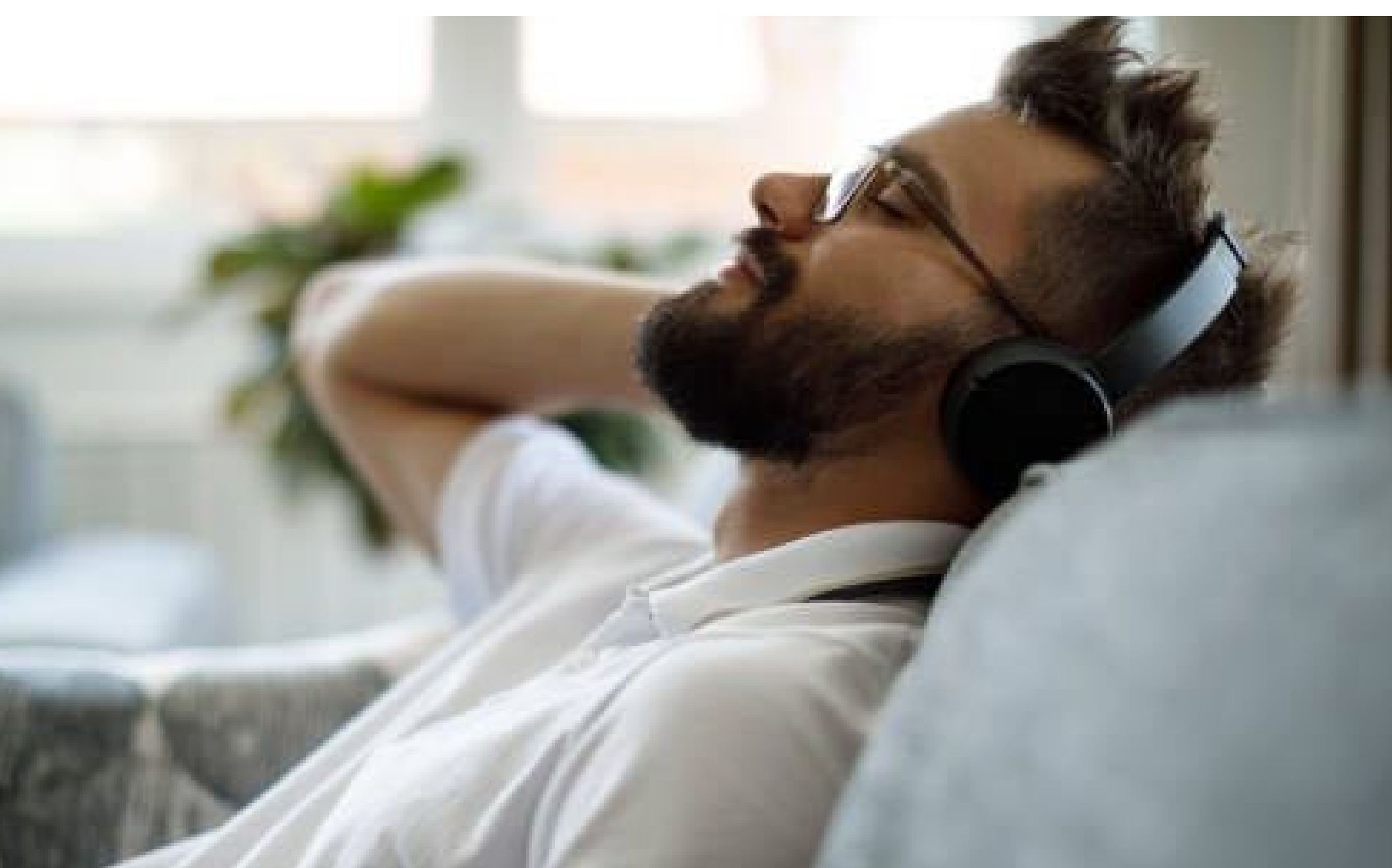
When I am listening to tunes (probably Vampire Weekend, Yung Bae or Weezer) I am going use my black Bowers Wilkins cans. Yeah pricey, but you pay for quality.



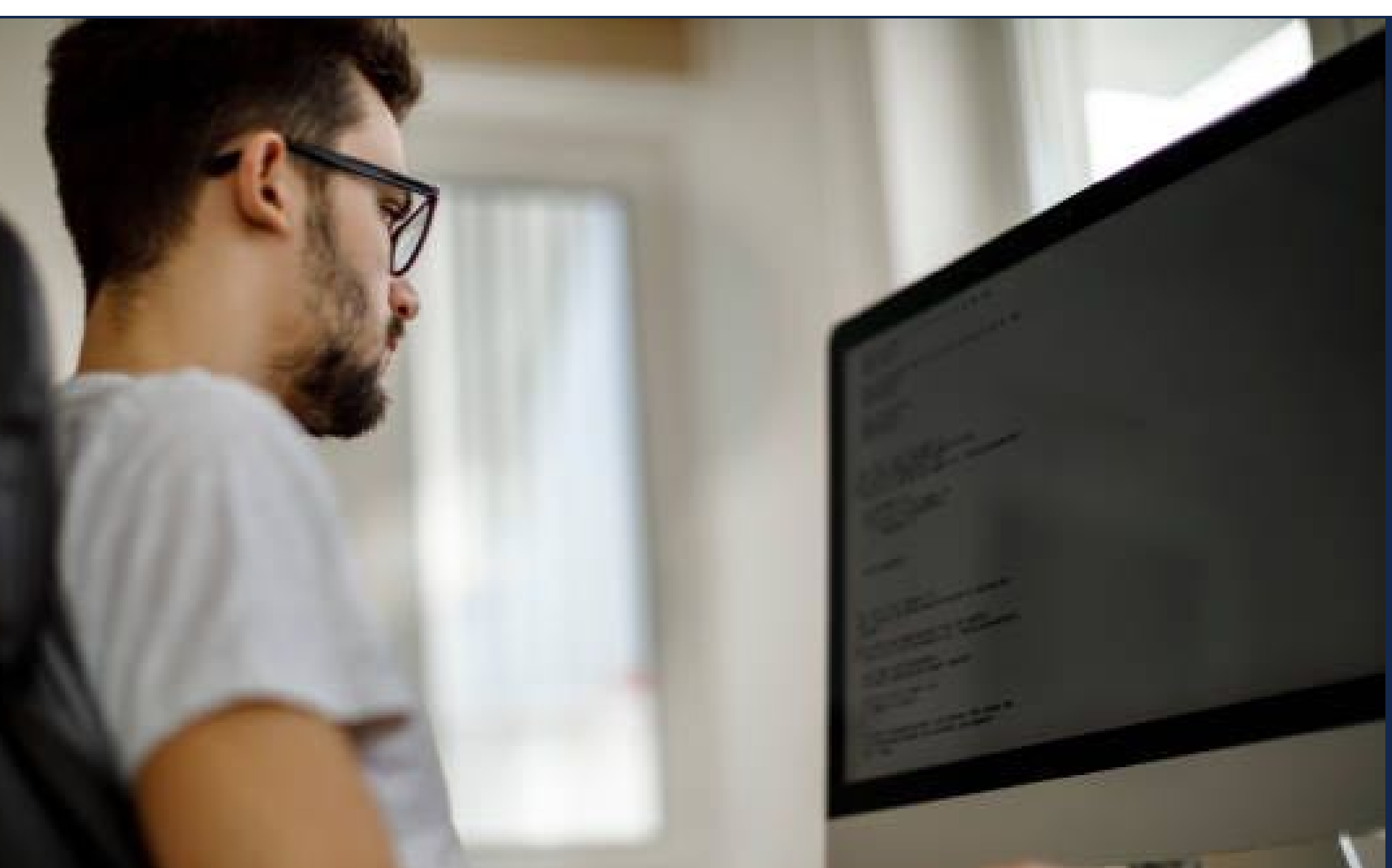
White phones are reserved for when I am watching videos and they match my room. I like to coordinate things and am a bit minimalist.



I am engaged in many things. I always gotta make enough time to work on my side hustle too! My headphone trading app won't build itself.

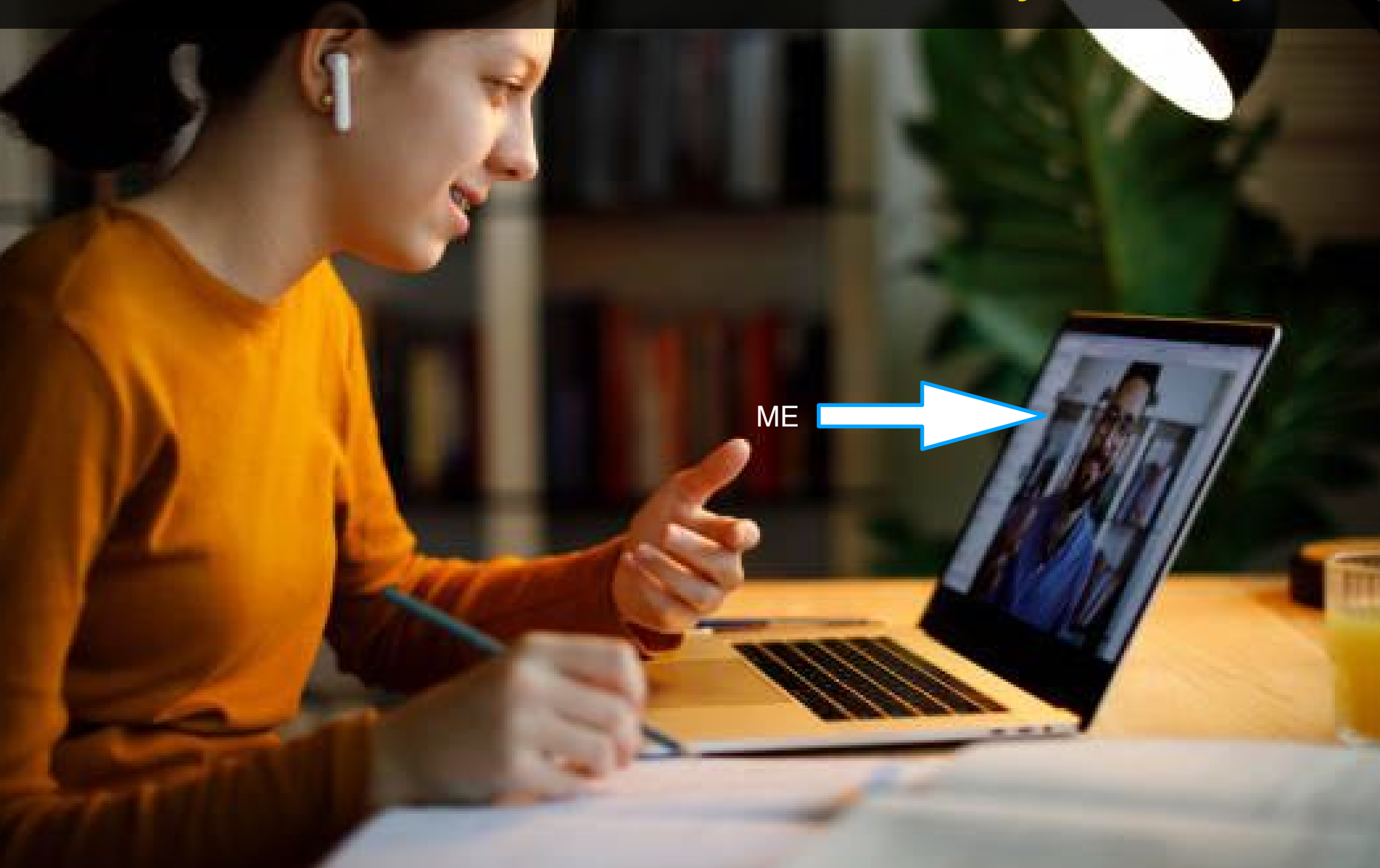


I am exhausted and yet barely anything got done.
Still, how am I supposed to work if I'm tired?



Ok that was a nice nap, so back to work!

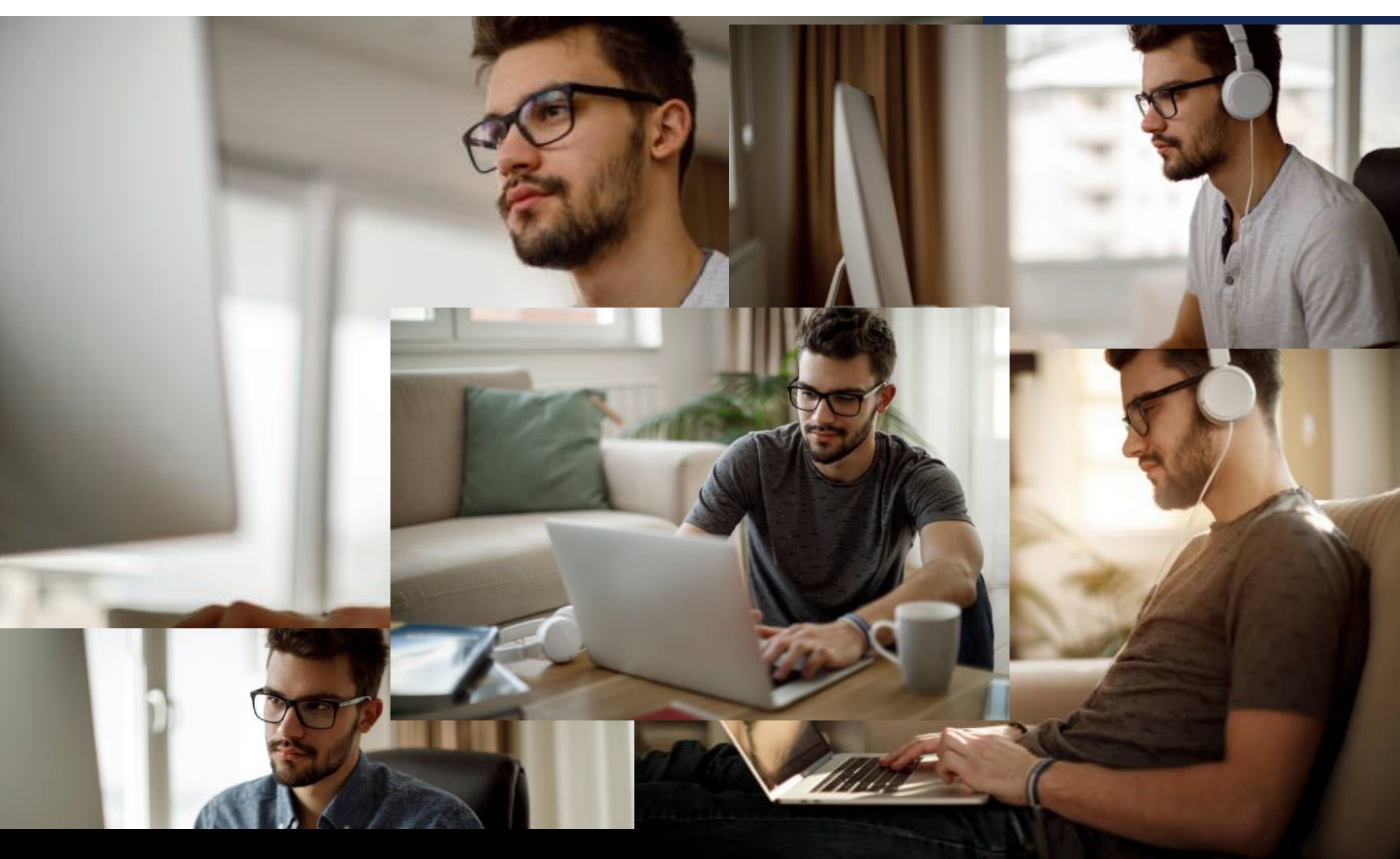
Oh no I almost forgot! I gotta help my sister first with her homework for Zoom University™ today.



ME



A got a Oculus Rift and since it is fully charged now, time for a couple of Beat Saber rounds.



So that's been my problem: It feels like everyday I just work...and work...and work... and get nothing done



My girlfriend Maria seems to have so much time to do the things she loves. She still plays guitar a lot and I want to know her secret.



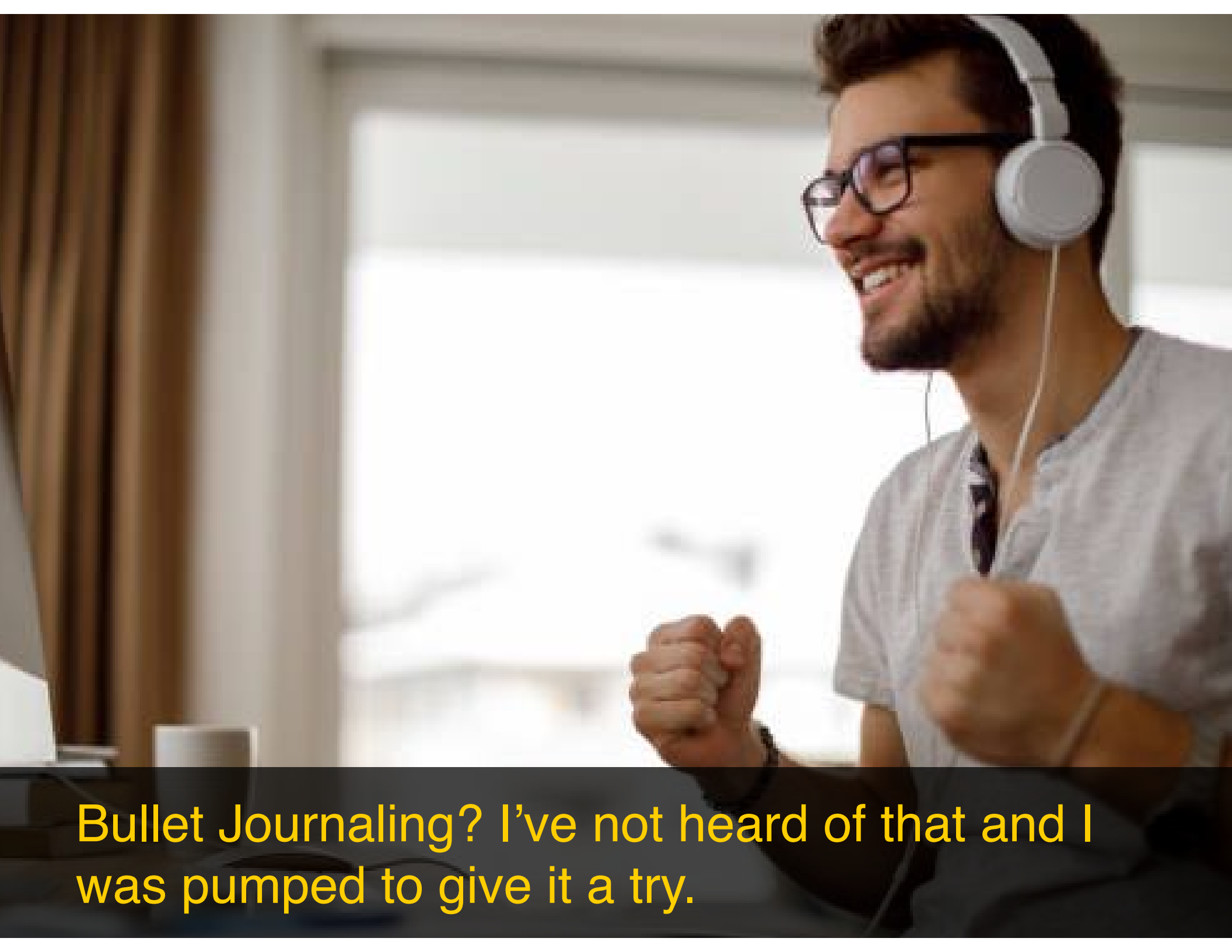
She is much more focused than me and seems to get everything done on time. I notice she's always writing things down.

So I decided to call her and ask about it

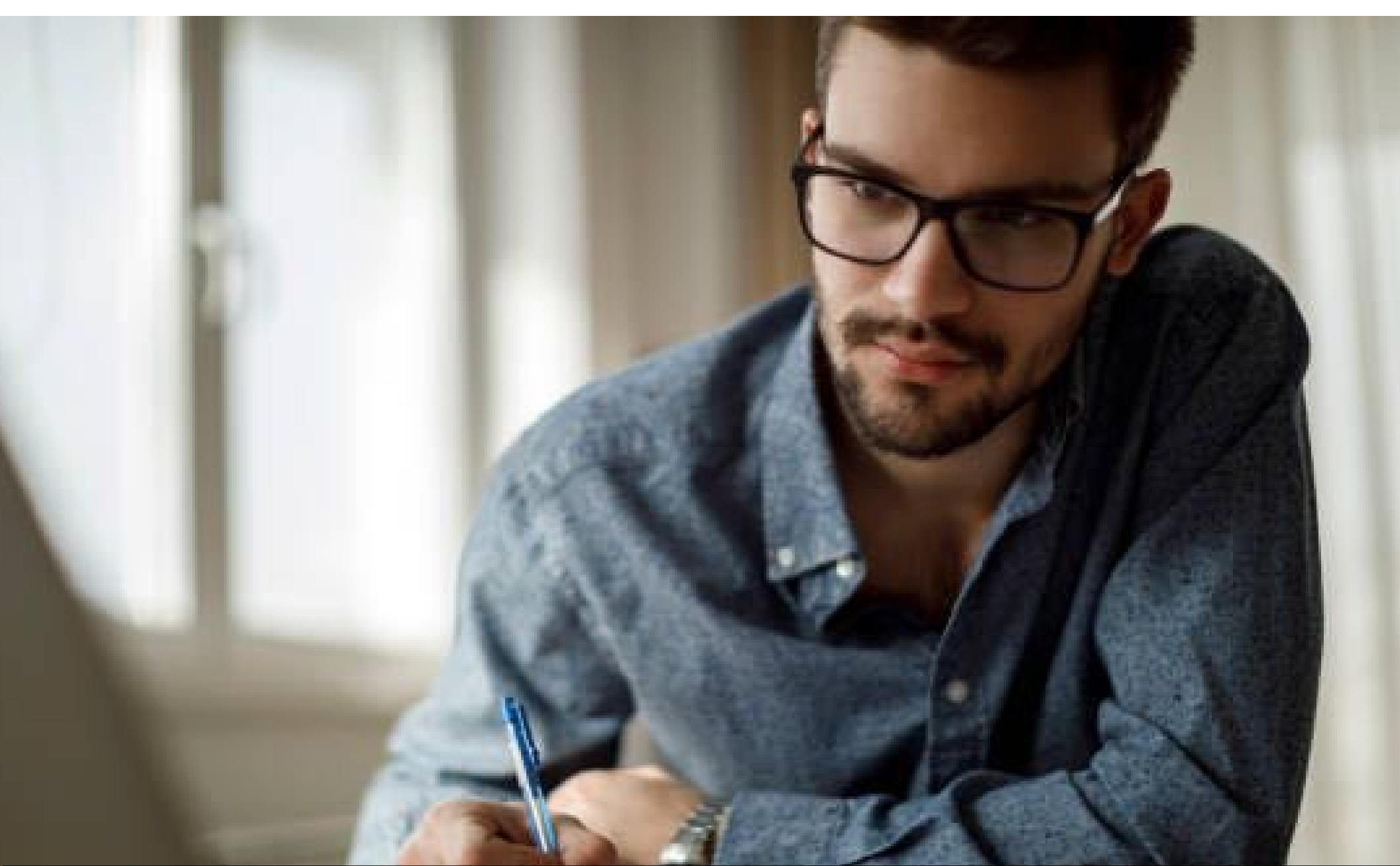




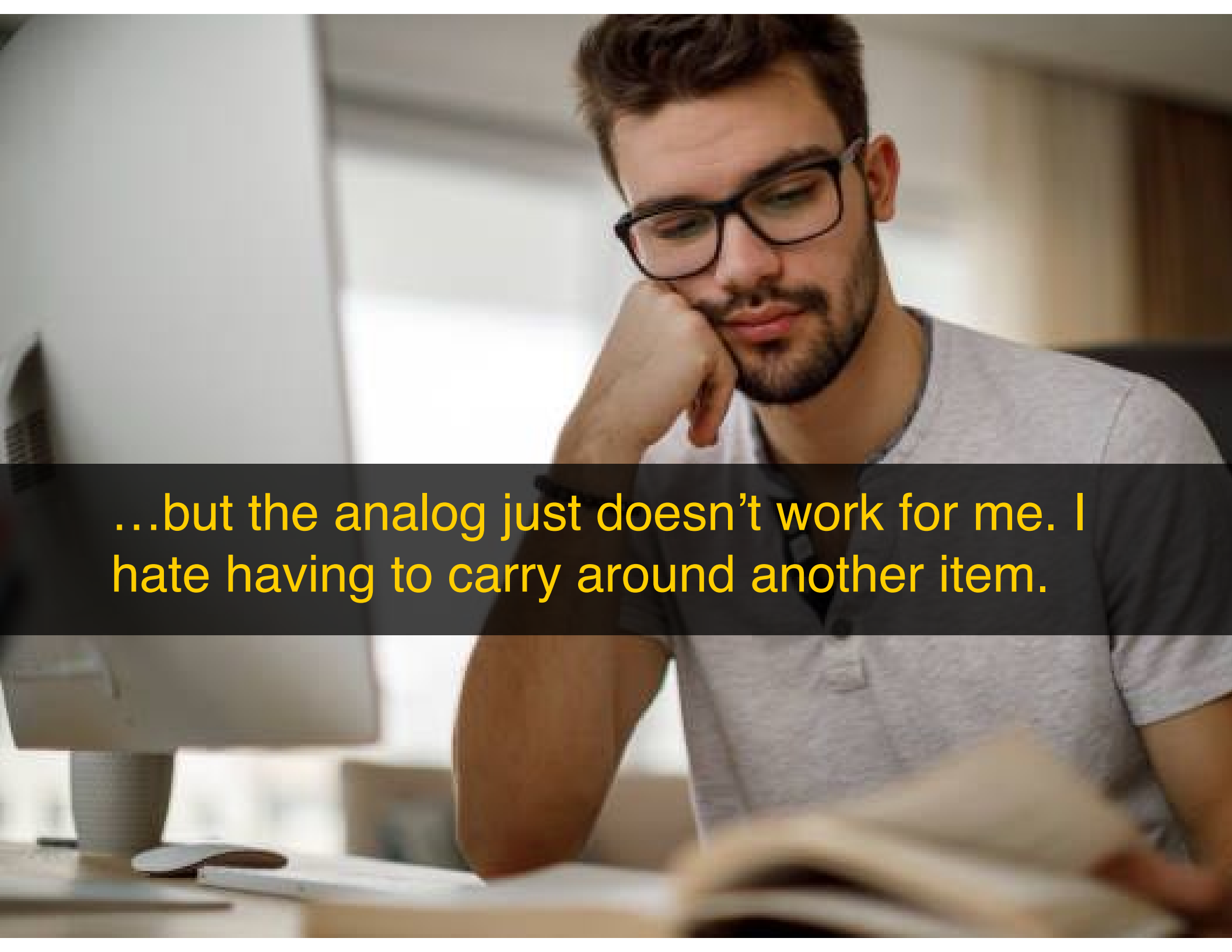
She tells me I should look into something called Bullet Journaling.



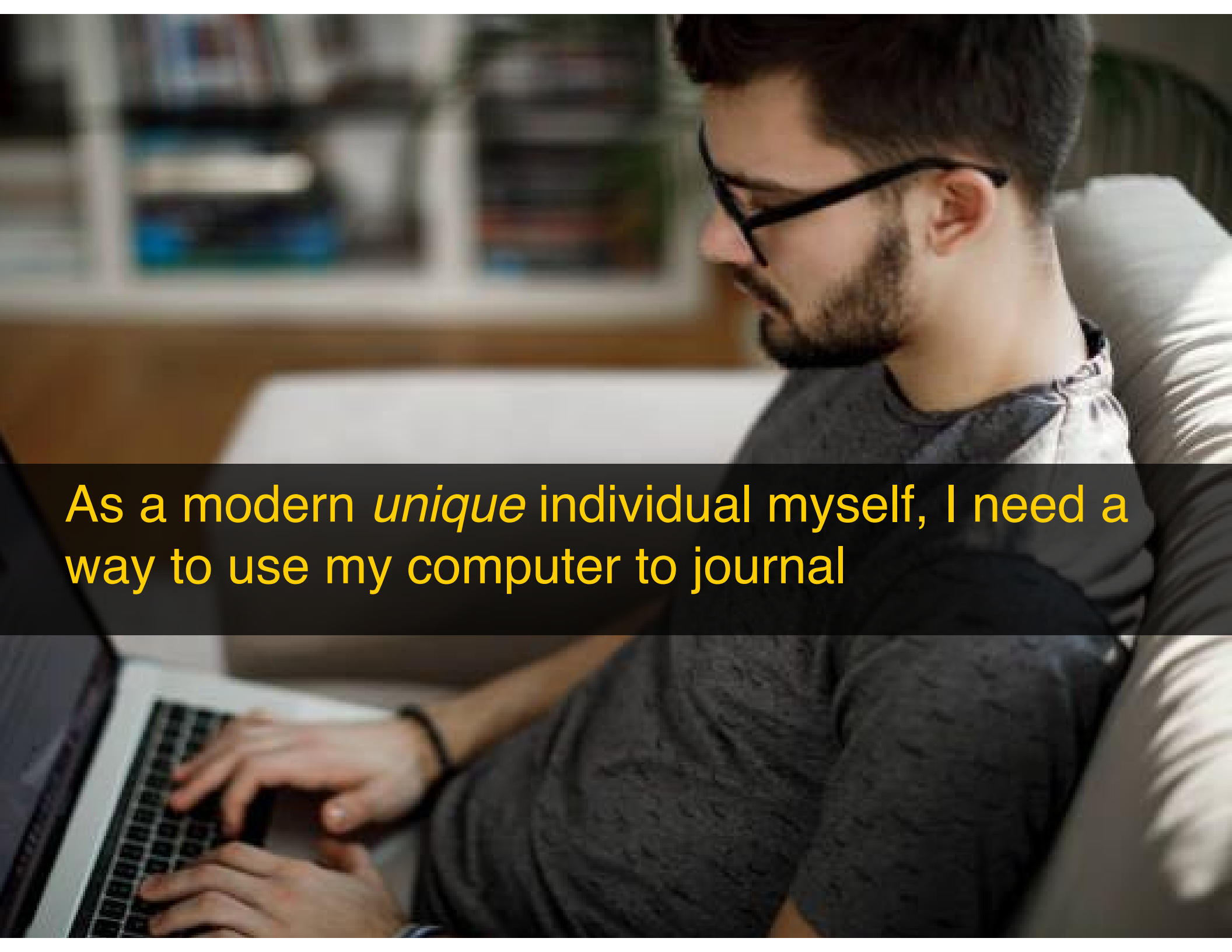
Bullet Journaling? I've not heard of that and I was pumped to give it a try.



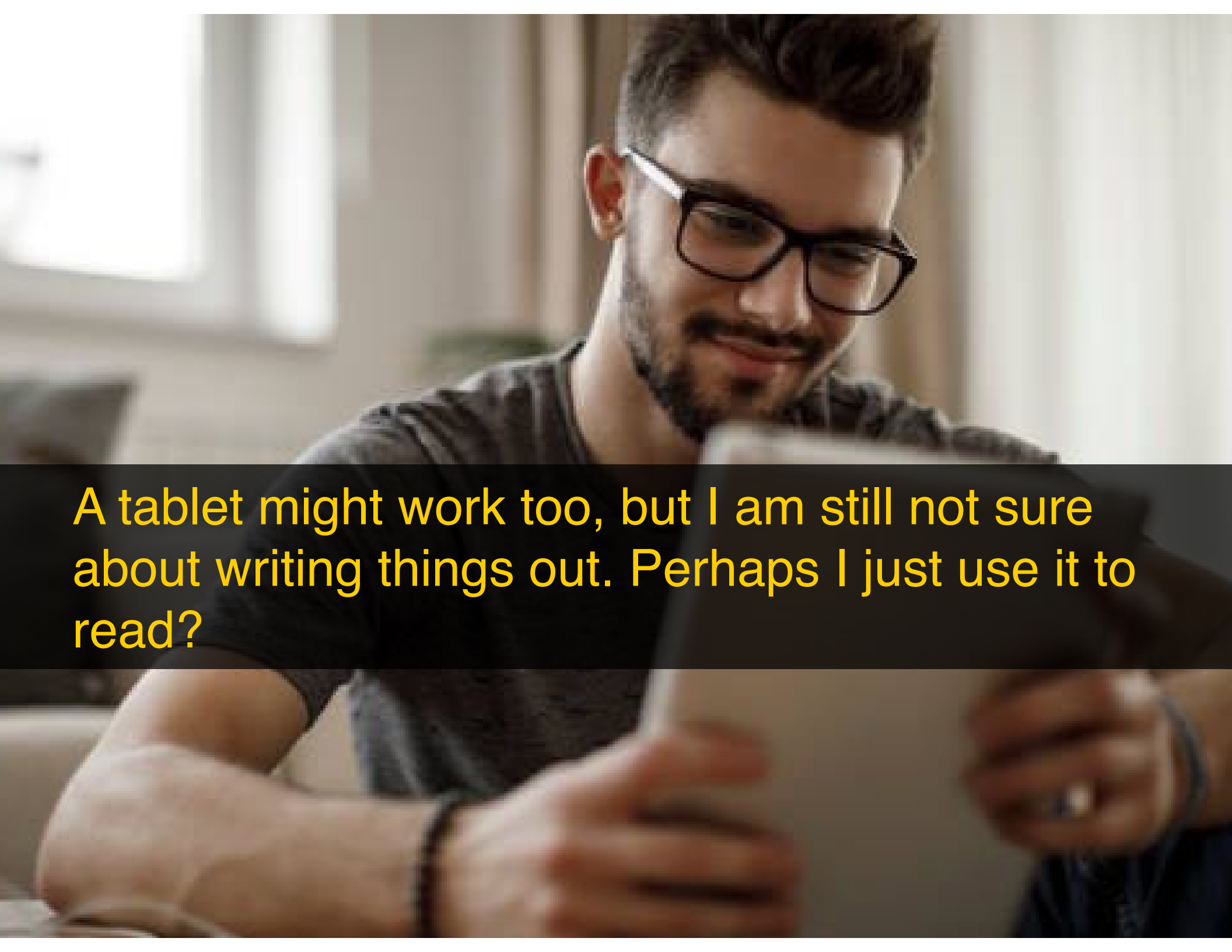
I tried it on paper and while it does seem useful...

A man with dark, curly hair, a beard, and black-rimmed glasses is sitting at a desk. He is resting his chin on his right hand, looking down with a thoughtful or slightly frustrated expression. He is wearing a light gray t-shirt. In the foreground, there is a computer monitor on the left and some papers or a book on the desk. The background is blurred, showing an office or home workspace with a window.

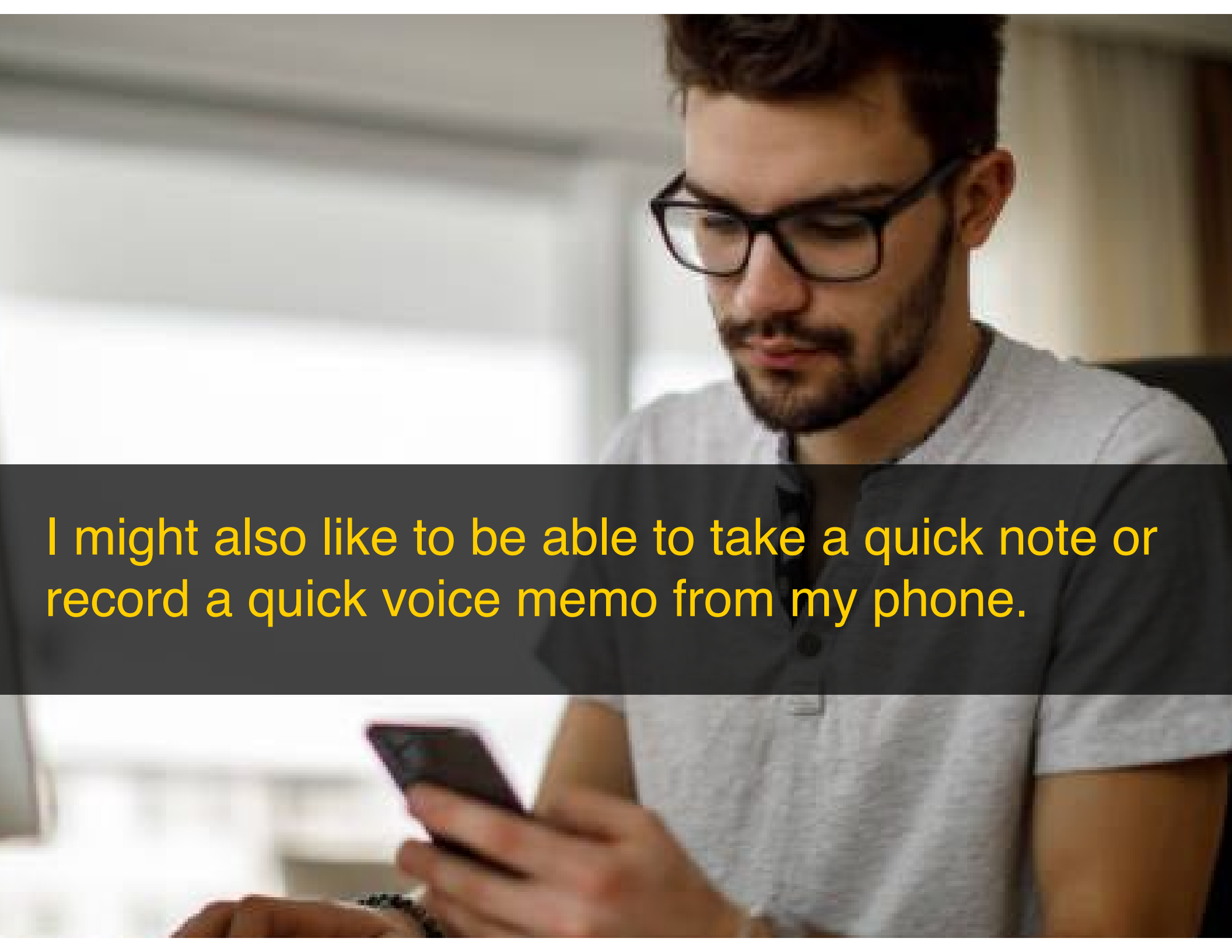
...but the analog just doesn't work for me. I hate having to carry around another item.

A man with dark hair, a beard, and black-rimmed glasses is sitting on a light-colored sofa. He is wearing a dark grey t-shirt and is focused on typing on a silver laptop. The background is a blurred living room with a bookshelf filled with books and a potted plant. A semi-transparent black banner with yellow text is overlaid across the middle of the image.

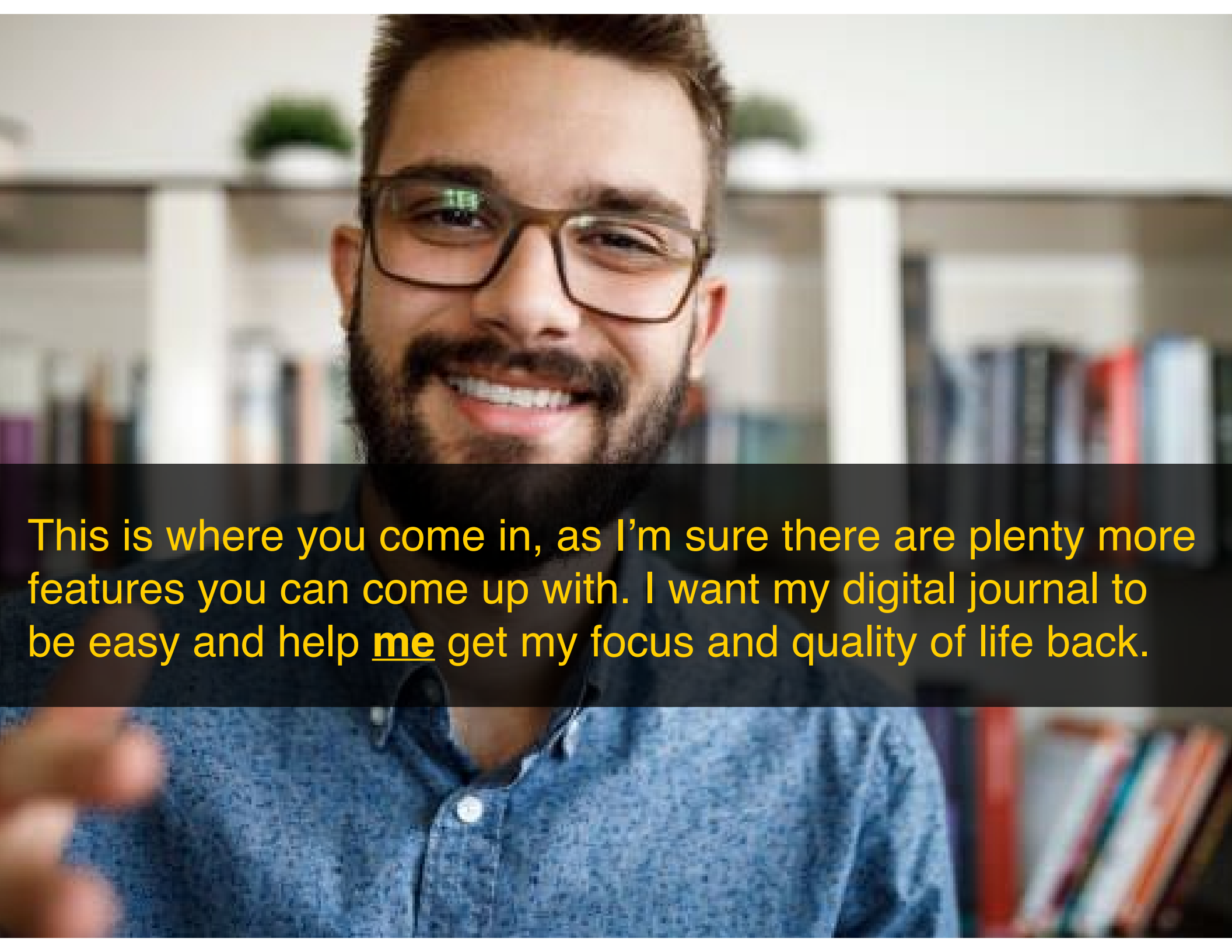
As a modern *unique* individual myself, I need a way to use my computer to journal

A man with dark hair, a beard, and black-rimmed glasses is looking down at a tablet computer. He is wearing a dark grey t-shirt. The background is a blurred indoor setting with a window. A semi-transparent dark grey banner is overlaid across the middle of the image, containing yellow text.

A tablet might work too, but I am still not sure about writing things out. Perhaps I just use it to read?



I might also like to be able to take a quick note or record a quick voice memo from my phone.

A close-up portrait of a man with a beard and glasses, smiling. He is wearing a blue button-down shirt. The background is a blurred library or bookstore with bookshelves. A semi-transparent black bar with yellow text is overlaid on the lower half of the image.

This is where you come in, as I'm sure there are plenty more features you can come up with. I want my digital journal to be easy and help me get my focus and quality of life back.



If you could figure out something to help me, I'll give you a big thumbs up just like this. Also I'll tell the Professor to give you an A+!



I know I'll be a lot happier if I can just get more focused. Please help!



Then, with your digital BuJo helping me and some of those sweet COVID Vaccines, I'll go back to my coffee coding HQ ready to win the Internet in 2022!

ALWAYS THINK OF CHAD

OUR STOCK PHOTO UCD HERO!

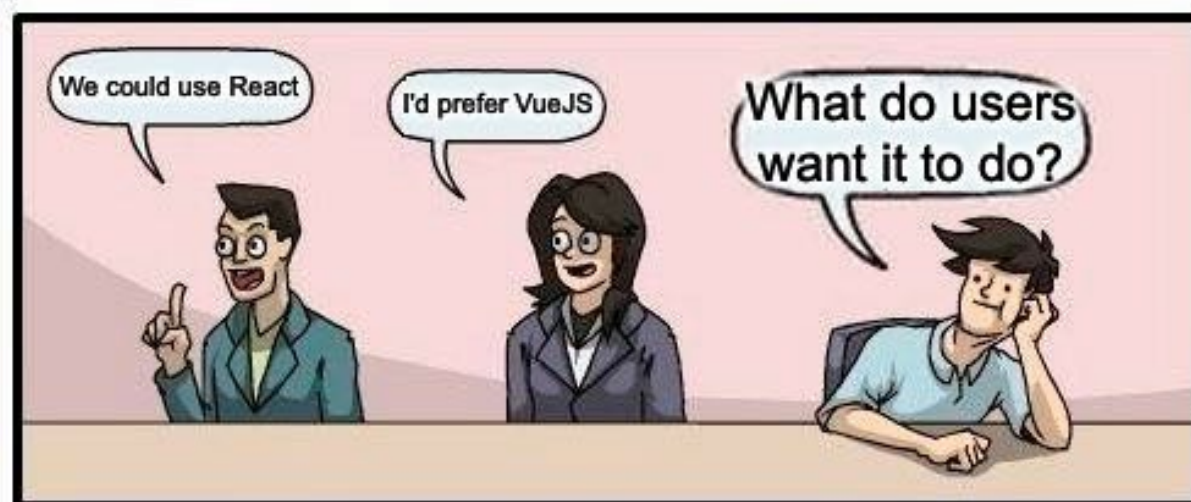
User Personification Summary

- Sure this was a goofy set of stock photos, but it was very purposefully built to tell a story of “Chad” and his issues.
- It is an effort to get you to consider visualizing and documenting a person’s life and what they might be facing
- We must move thinking from a code to be built to a job to be done for a user by the code you might build.

Reminder to Act Your User Values

Empathetic Actions > Empathetic Feelings

- Few people are going to say we are anti-users. We know users matter, they pay our bills in some sense
- Few of us pay no attention the idea of end users and their situations
- However, our feelings towards users aren't as important as our actions towards them
 - Remember talking about accessibility and agreeing it is a good aspiration is a feeling. Trying to add some keyboard navigation or degradability under less ideal conditions is an action.
 - Disrespecting users by not bothering to ponder their needs deeply or even talk to them is an action of omission. Even if you feel you are user centered at your heart, your actual actions do not back this up when you jump into solutions with tech before deeply understanding the problems of your users.



Take Action Instead

- Deeply articulate users and their problems first
- Choose appropriate solutions second
- Then make your plan
- Finally execute it and iterate as you encounter changes

Concern: This is SE, Not HCI or CogSci!

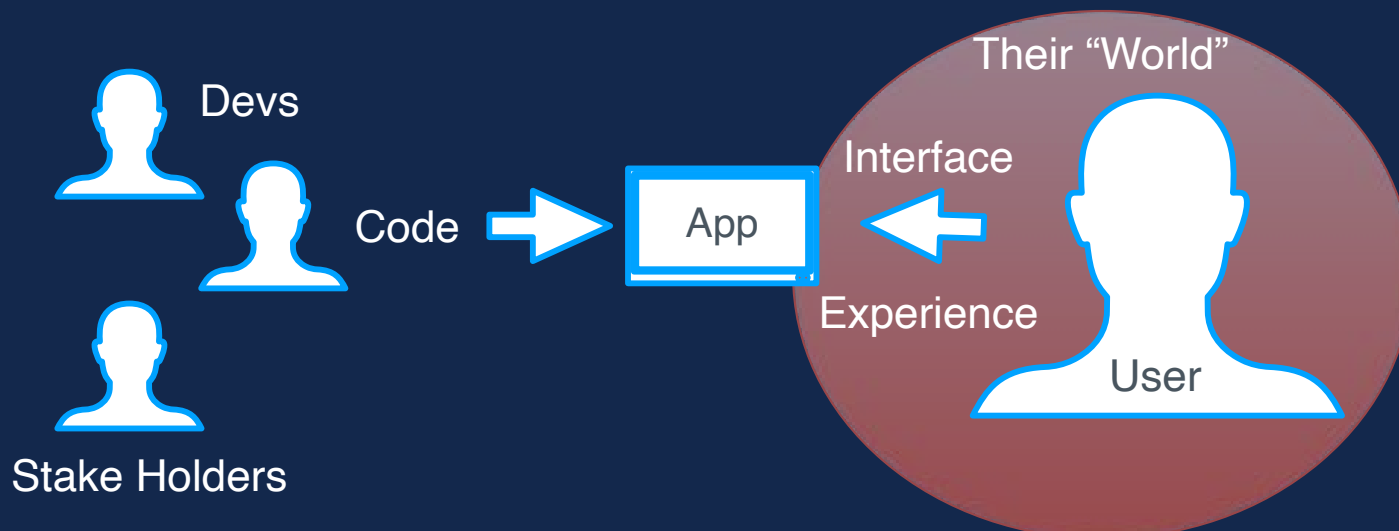
- Actually the Agile Manifesto cites users A LOT, so apparently you think that is wrong?
- Fundamentally EVERYONE building a product needs to care about user experience as it all for users and without those users we are nothing.
- TL;DR - Avoid user centered thinking at your project, product and career peril.



Design Process Basics

Copernican Development

- Move the user to the center of the system as opposed to the developer at the center of the system
- Code outwards is a developer outward philosophy
- Interface inwards is a users inward philosophy



User Centered Design (UCD)

The UCD process is one which the needs, wants and limitations of end users of a product, service or system are given extensive attention and consideration at each stage of the design and development process



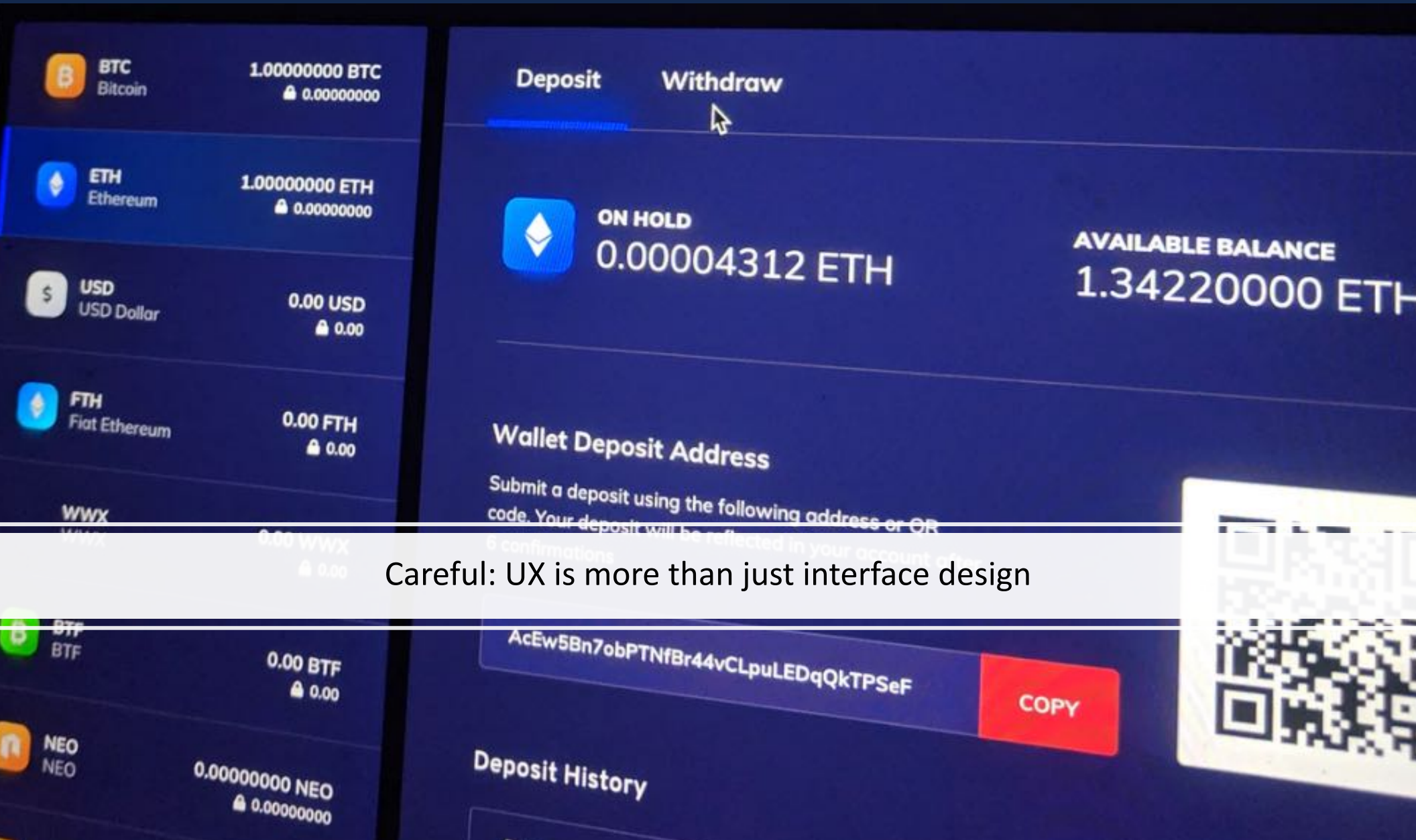
Value Centered Design vs User Centered Design?

- Similar to the DX - UX See-Saw consider the idea of the value intersection

Value

A value centered design philosophy focuses on the benefit balance we should aim for between individual users and organizations who build the software for them

Beware though Total UX > UI Pixels



Think Broadly

“User experience encompasses all aspects of the end-user’s interaction with the company, its services, and its products.”

Don Norman



Determining what users want is the very heart of UX

Santa – Expert Level Experience Specialist

Make Sure You Capture the Whole Experience

- Experiences are personal and your design artifacts will need to capture this
- Perception of an experience can be situational and cumulative having been effected by items (roller coaster after large meal or not)
- However like any visualization a diagram, map, etc. cannot show all details.
 - If we show everything we'd have the extreme complexity of “reality” what we want is a useful simplification of that complexity into something everyone understands
 - It is important to note both what we choose to show and what we don't. Omission is editorial and a design decision you are making.

While Utility + Usability = Usefulness



Jus that isn't usually enough

Jobs to be Done

- A user in some sense hires your product to full fill some “job” or need they have.
 - Job is an easy short-hand for what a user really wants to accomplish under a particular circumstance
 - Circumstances often outweigh user characteristics - two very different people can have under a certain circumstance have a shared job to be done.
- Does “Chad” believe that your solution will solve his “job to be done”?
- Your Alignment Diagram(s) and other artifacts need to tell that story

The Moment of Truth

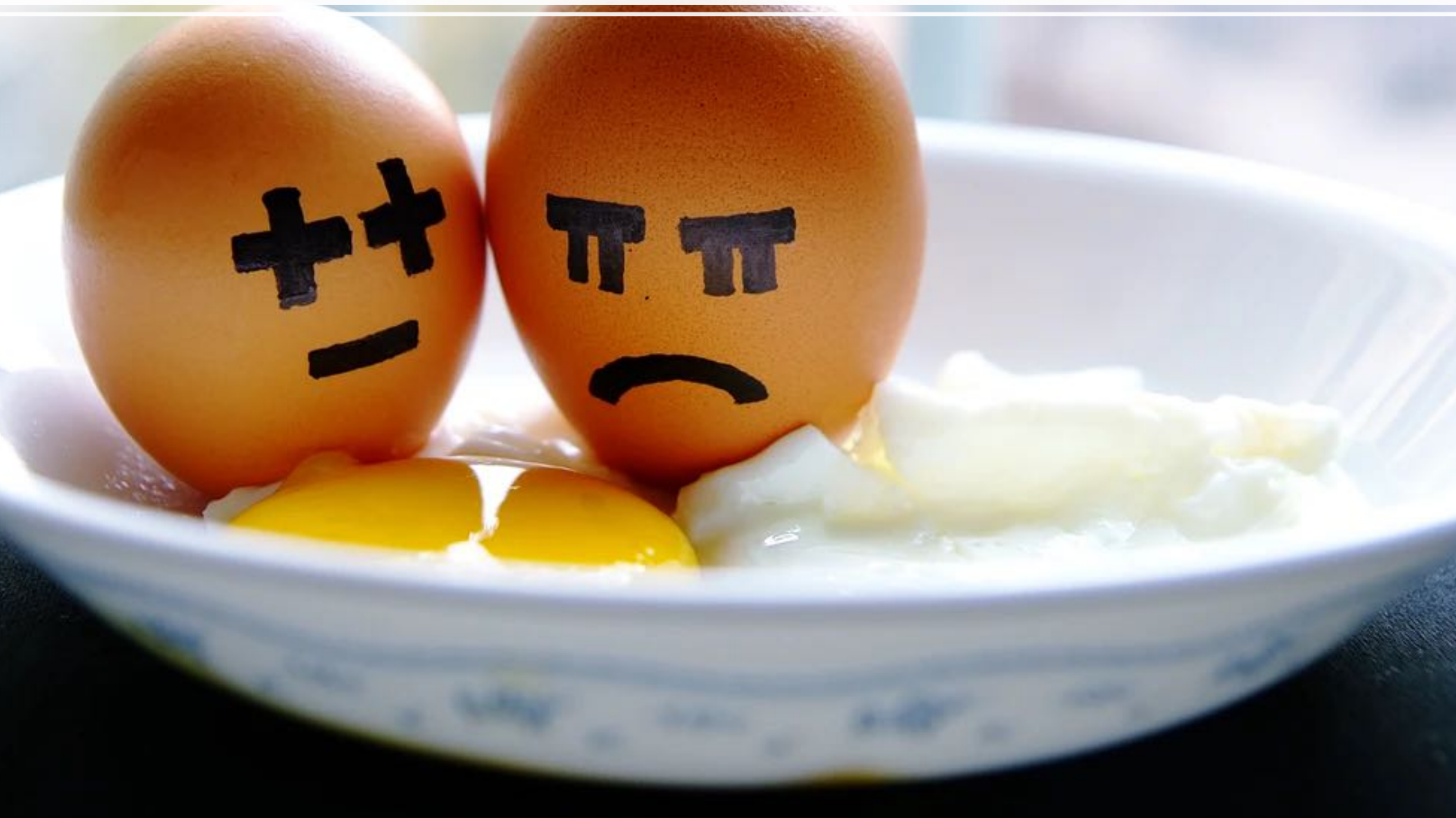
- Assuming you are on the user's road in their problem journey there is a key **moment of truth** for the product is when Chad the user's "meet cute" with your product goes one way or an other. Will Chad give your < blank > a chance or will he move on to the next experience in his life?

A flood of data and emotion floods Chad's system. Does < blank > fit with his life? Remember what Maria said about these things. What about < alternative blank > would what be a better choice? Does he really want < blank > or is he bored and more interested in something else?

- We wrote this all out and it likely is quite a bit more involved with this. Subtle things including mood might effect if Chad is gonna use your app.

TL;DR - Think beyond feature utility think about user feelings and outcomes

Enjoyment > Usefulness?



First don't bother them, before you try to delight them

**Ken Kocienda**
@kocienda

When I design software, I think a lot about people who don't care about computers. They just want things to work with a minimum of fuss.

7:15 AM · Sep 20, 2022 · Twitter Web App

17 Retweets 5 Quote Tweets 220 Likes



**Tweet your reply**Reply

**Ken Kocienda** @kocienda · 3h
Replying to @kocienda
I expect most people don't look to casual interactions with their gadgets to be fun, especially since it's often so frustrating to do simple things.



**Ken Kocienda** @kocienda · 3h
I also expect designers and engineers to employ all the care and craft they can muster to make the best work possible—but always with this idea in mind too: a little whimsy goes a long way.



Watch
Out for
Missing
Variables!



Capture the Whole Thing in Context and Our Time Period

Culture and
context
really
matters!



Some solutions solve problems only in a certain context or during some period of time.

Fight User FUD with Your Software DUF?

- User problems are often driven by FUD
- Solutions interestingly can be driven by DUF!



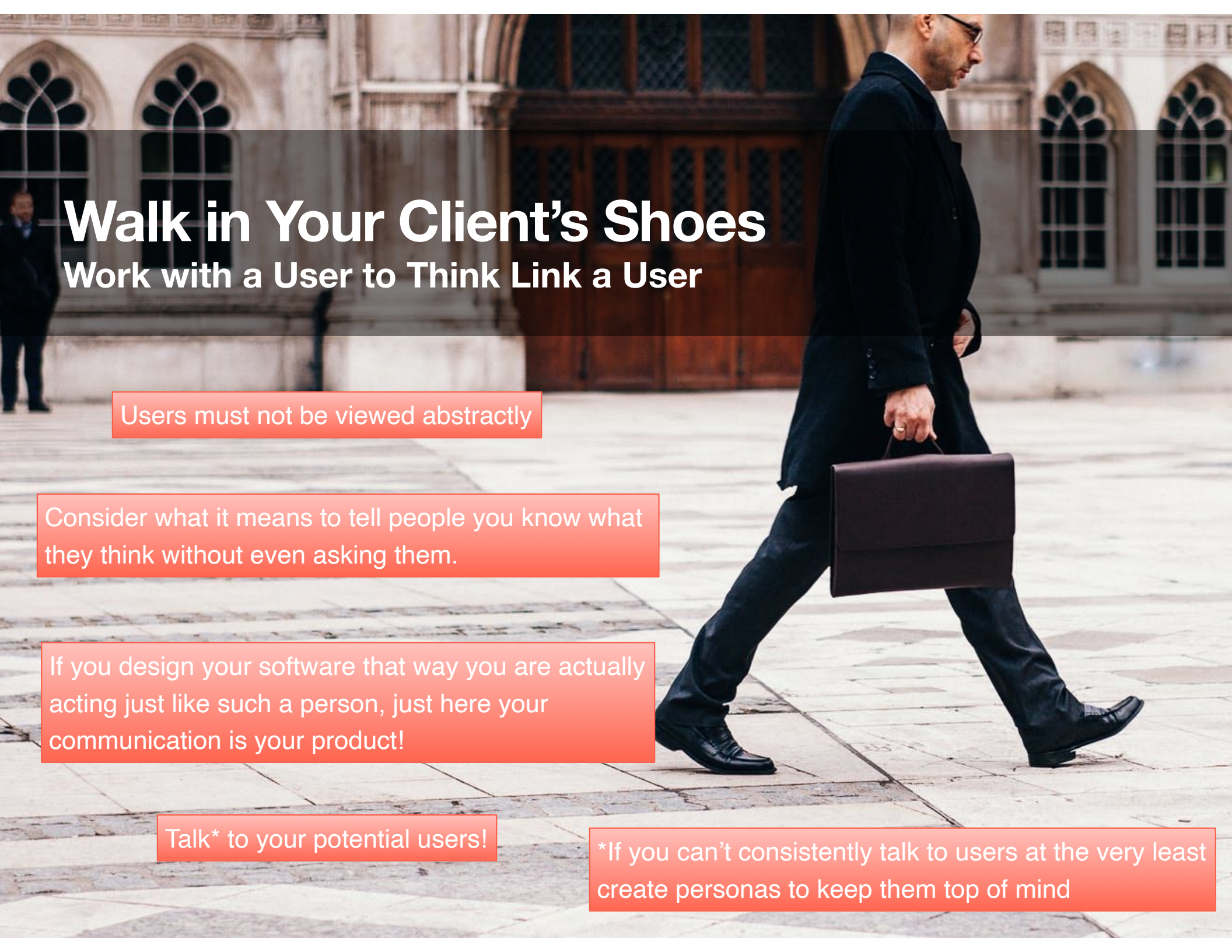
None of this is new. True in 1997, True in 2097



More here on full exchange <https://www.inc.com/justin-bariso/20-years-ago-steve-jobs-demonstrated-the-perfect-w.html>

Begin Building Outside In Alignment

- A goal of a good design effort is to bring alignment between internal developer perceptions and external users needs.
- Only after devs form a clear understanding of users and empathize with the problem they (the users) are trying to solve can we safely move from defining a problem to creating a solution
- As a reminder often devs move to a solution before the problem is understood or at least understood the way the user (as opposed to the dev) actually views that problem.
 - Note: We repeat this is over and over because it is so common of a mistake it almost feels as if is accepted as somehow not a mistake.



Walk in Your Client's Shoes

Work with a User to Think Link a User

Users must not be viewed abstractly

Consider what it means to tell people you know what they think without even asking them.

If you design your software that way you are actually acting just like such a person, just here your communication is your product!

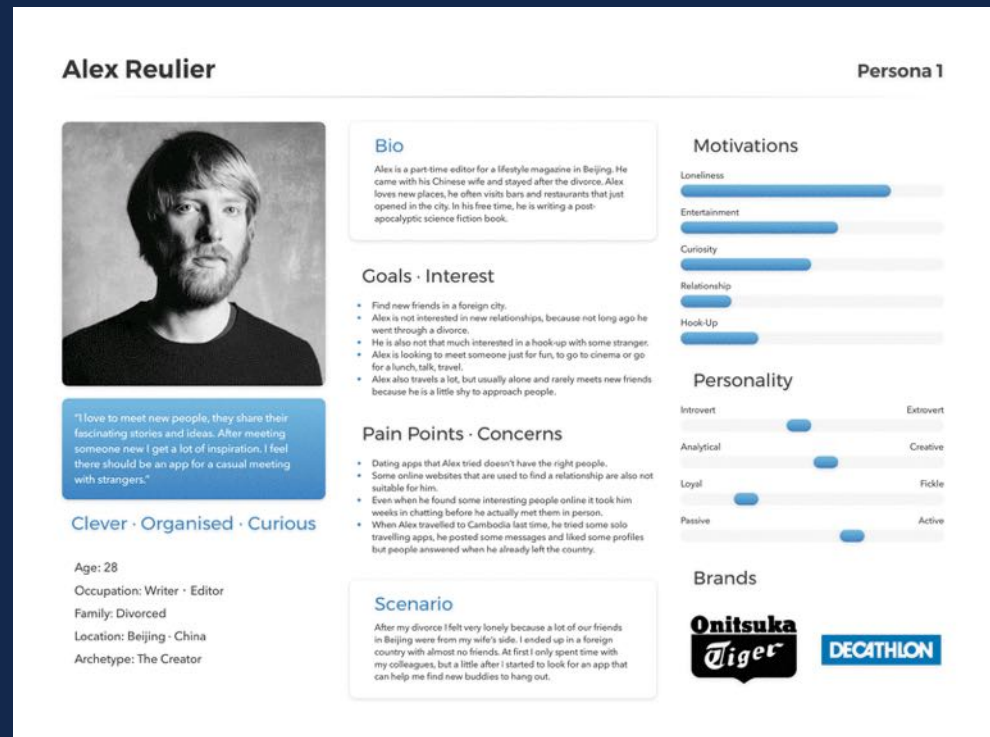
Talk* to your potential users!

*If you can't consistently talk to users at the very least create personas to keep them top of mind

Project Artifact: Personas

Personas

“A persona, in user-centered design and marketing is a fictional character created to represent a user type that might use a site, brand, or product in a similar way.”



<https://www.interaction-design.org/literature/article/personas-why-and-how-you-should-use-them>

<https://www.usability.gov/how-to-and-tools/methods/personas.html>

Personas for SEs

- While we aren't in a marketing / HCI role without personas we are building features in space not for people
- We must have personas first to understand our users and their possible problems and the solutions we might provide.
- The format of personas will vary but they should make the “person” real enough and capture details that should be kept in mind as the software is built.

Make Persona's Real Enough

- A persona should provide enough details about the “person” described that they seem real
- Avoid movie stars, cartoons, etc.
- Movies can be instructive as the texture and back story of a well designed character in a movie is a common trait of successful stories that viewers engage with



Julia's Cafe / Ideal User Persona

Janelle Robinson

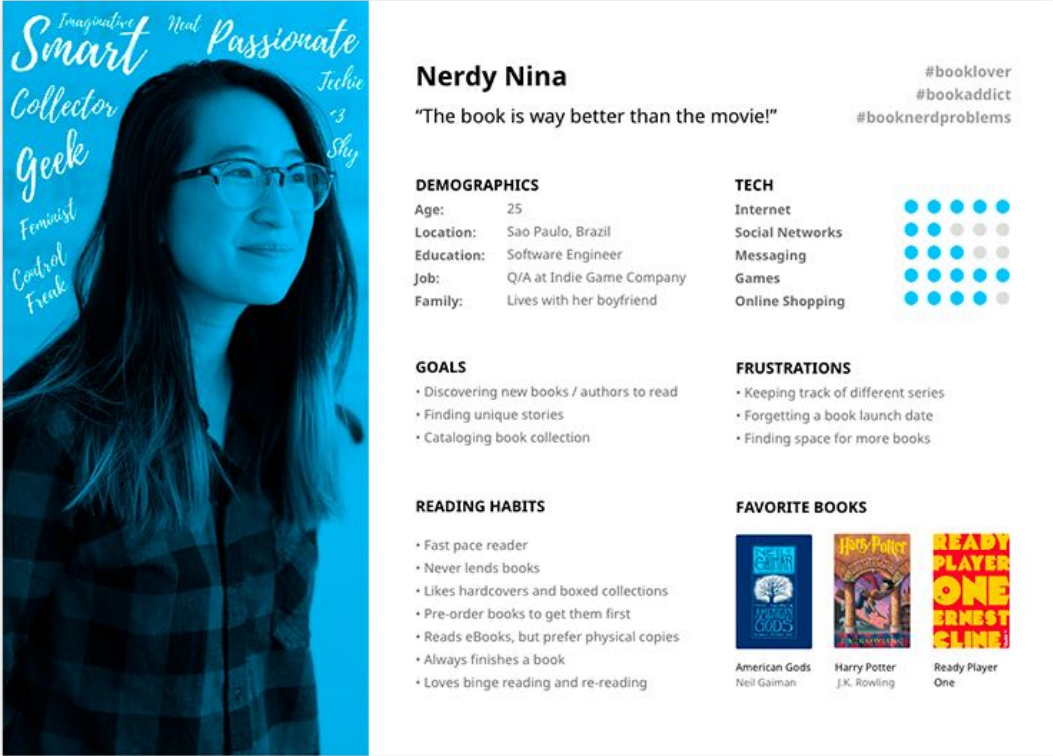
Age: 26
Status: Undergraduate

Customer Profile: A busy PhD Student who needs a quiet place to study and read without distractions. She spends a lot of time on campus, refuels often and is a major coffee lover. She is the ideal customer for Julia's Cafe. She wants to receive quick and professional service; order online from her smartphone to avoid lineups, and not deal with over-conversational staff members.

Motivations	Frustrations
• High quality, strong coffee. • Quiet atmosphere for studying. • On or close to the college campus.	• Hates waiting too long for her coffee. • Doesn't want to be bothered by the staff while studying. • Actually relies on strong coffee to help her concentrate later in the day.
Goals	Favorite Coffee Beans
• She needs information on the type of coffee a cafe serves, in order to determine the quality. • She wants to find the menu and the daily specials as quickly and easily as possible. • She wants to order with her smartphone, avoid the usual lineup, and experience minimal distractions.	• Ethiopian Yirgacheffe Coffee • Sumatra Mandheling Coffee • Ethiopia Charbanta Natural

Be Careful with Shorthands and Stereotypes

- Alliteration is fun and “Nerdy Nina” might be a nice short hand name, but did we diminish the persona this way?
- It’s a common practice, but be careful with it.



Imaginative Smart Nerdy Passionate Geek Feminist Control Freak

Nerdy Nina
“The book is way better than the movie!”
#booklover #bookaddict #booknerdproblems

DEMOGRAPHICS
Age: 25
Location: Sao Paulo, Brazil
Education: Software Engineer
Job: Q/A at Indie Game Company
Family: Lives with her boyfriend

TECH
Internet
Social Networks
Messaging
Games
Online Shopping

GOALS
• Discovering new books / authors to read
• Finding unique stories
• Cataloging book collection

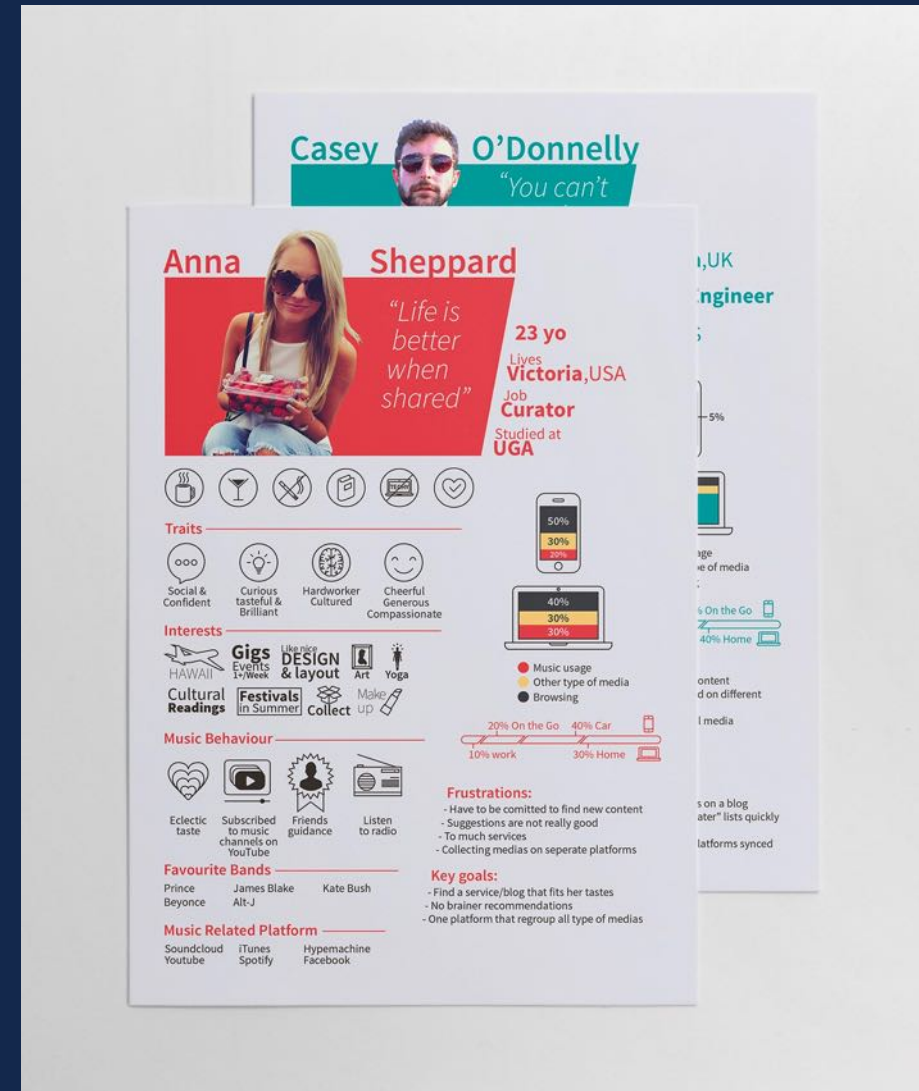
FRUSTRATIONS
• Keeping track of different series
• Forgetting a book launch date
• Finding space for more books

READING HABITS
• Fast pace reader
• Never lends books
• Likes hardcovers and boxed collections
• Pre-order books to get them first
• Reads eBooks, but prefer physical copies
• Always finishes a book
• Loves binge reading and re-reading

FAVORITE BOOKS
 American Gods
Neil Gaiman
 Harry Potter
J.K. Rowling
 Ready Player One
Ernest Cline

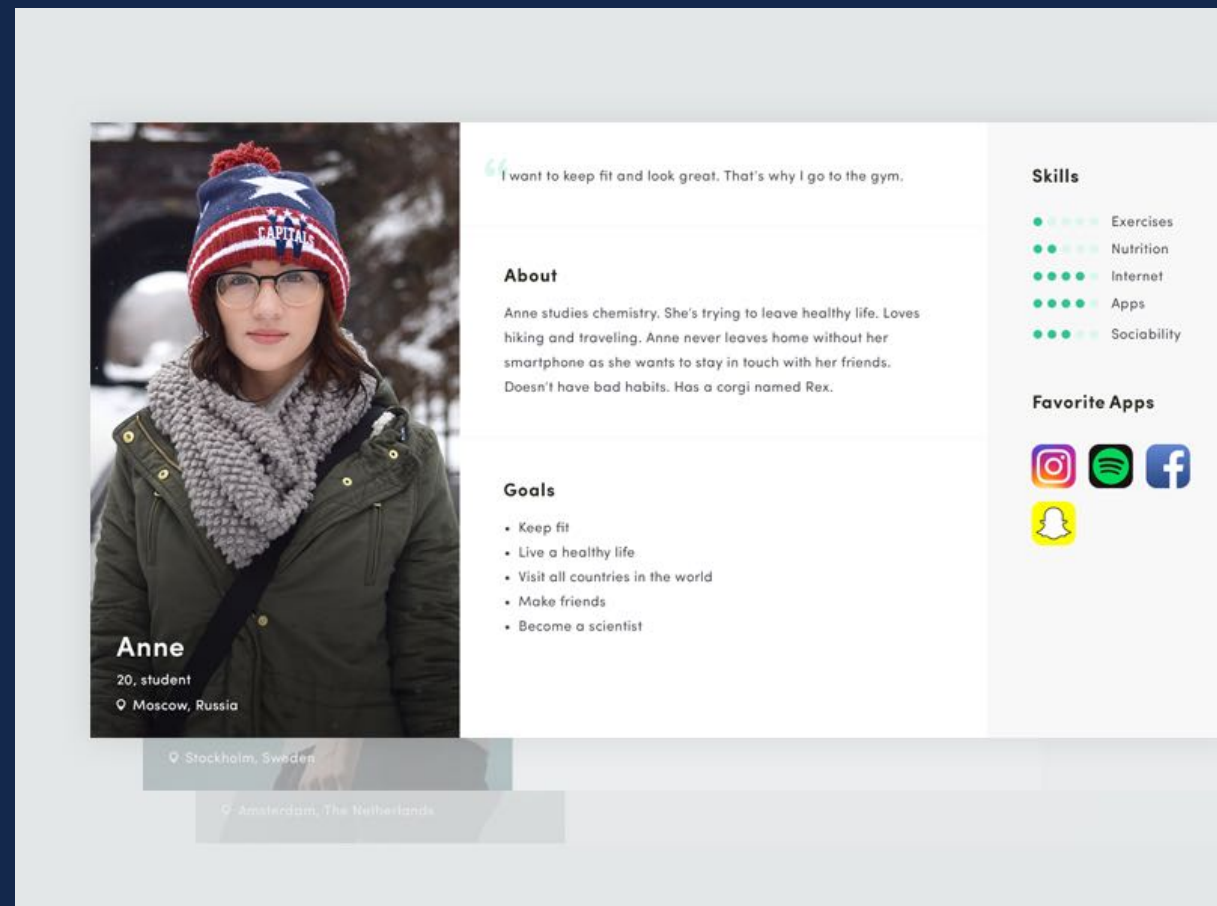
Use a Consistent Format

- Whatever format you end up using keep it consistent
- Make sure it is broad enough in what it captures
- Make it aesthetically appropriate to the group it is servicing and to show the care in its development



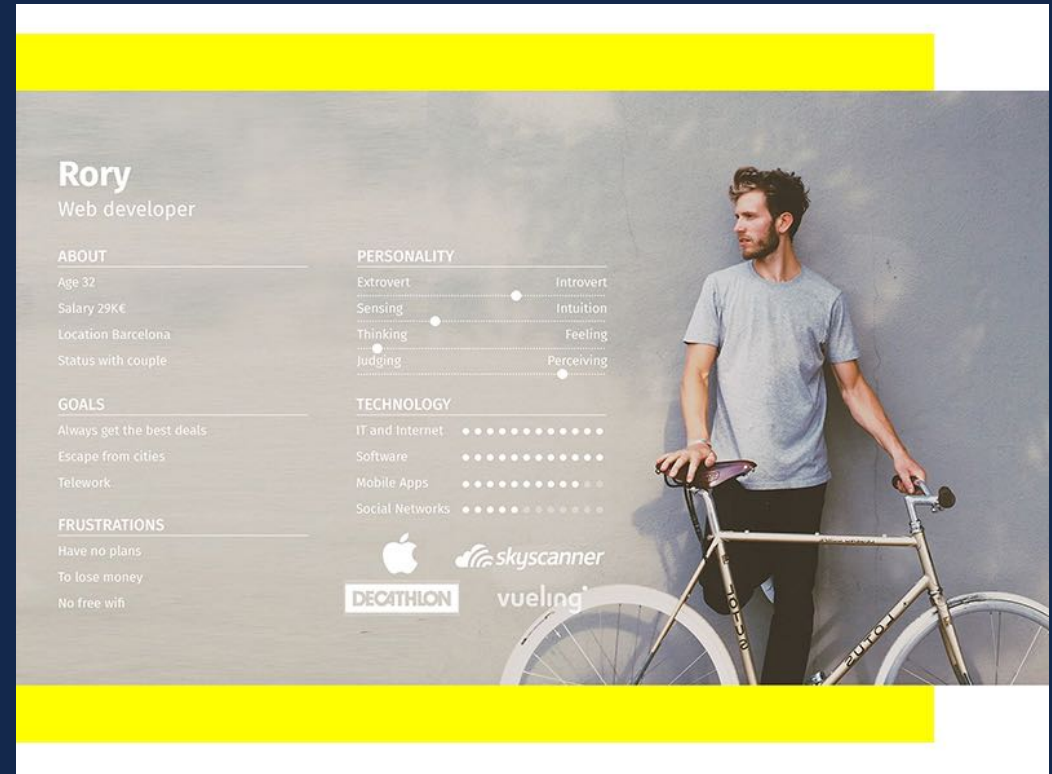
Persona's Should Acknowledge a World Beyond

- Your product isn't the user's whole universe
- In fact your product may be but a small piece in some broader mission or goal they have
- Make sure that broader goals and problems are presented in personas so you keep people "real"



Consider Ranges in Personas

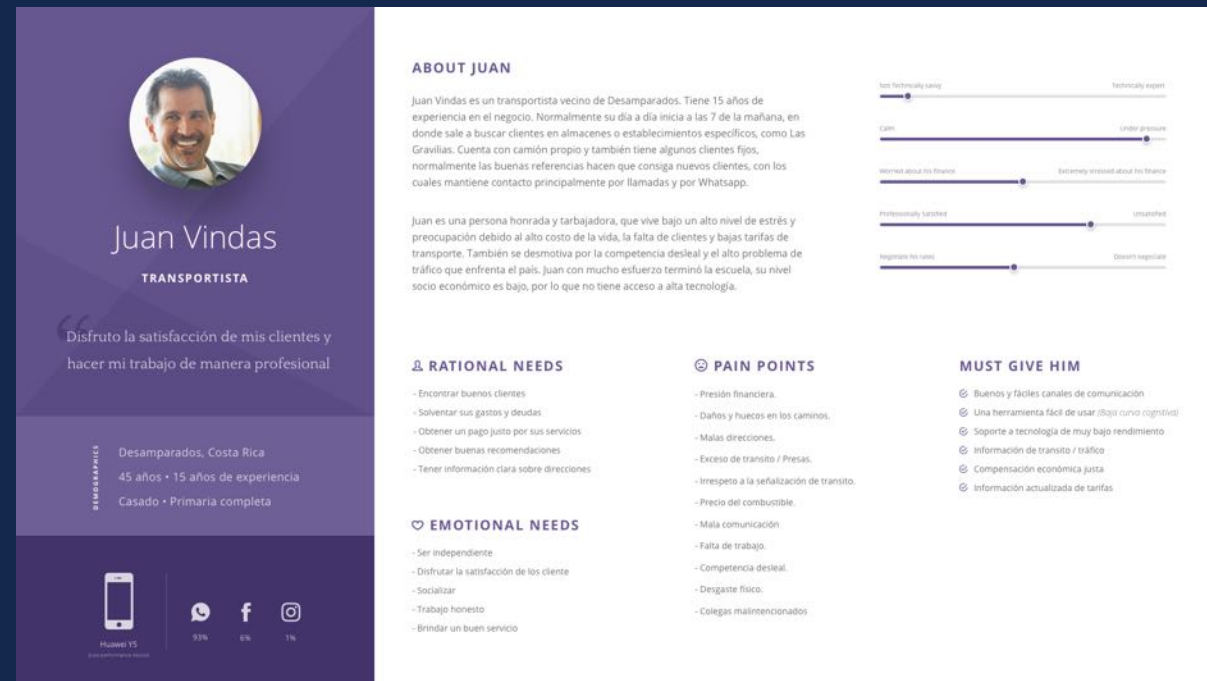
- Given that users of your product may be varied in their characteristics consider ranges or scales to give a better sense of user diversity
- Ranges might be useful for
 - Feelings
 - Interests
 - Skills
 - Knowledge Levels



Source: <https://dribbble.com/shots/4283788-Personas-cards>

Persona's Should Address Logic and Emotion

- Personas should only be about relating a person to features or use cases.
- Consider how people feel about the problem and solutions more broadly.
- What are their pains and desires that your solution might address?



Source: <https://dribbble.com/shots/3628471-FREEBIE-User-Persona-Template>

Beyond Actions are Thoughts and Feelings

- Commonly when devs think about user's it is about their actions. What feature did they use? What button did they press?
- Actions only is an incomplete description. We must consider the thoughts and feelings of the user before, during and after the action performed
- Avoid the second parts and your system only has utility, if you consider the others it may become useful or even enjoyable.

Project Artifact: Journey Maps

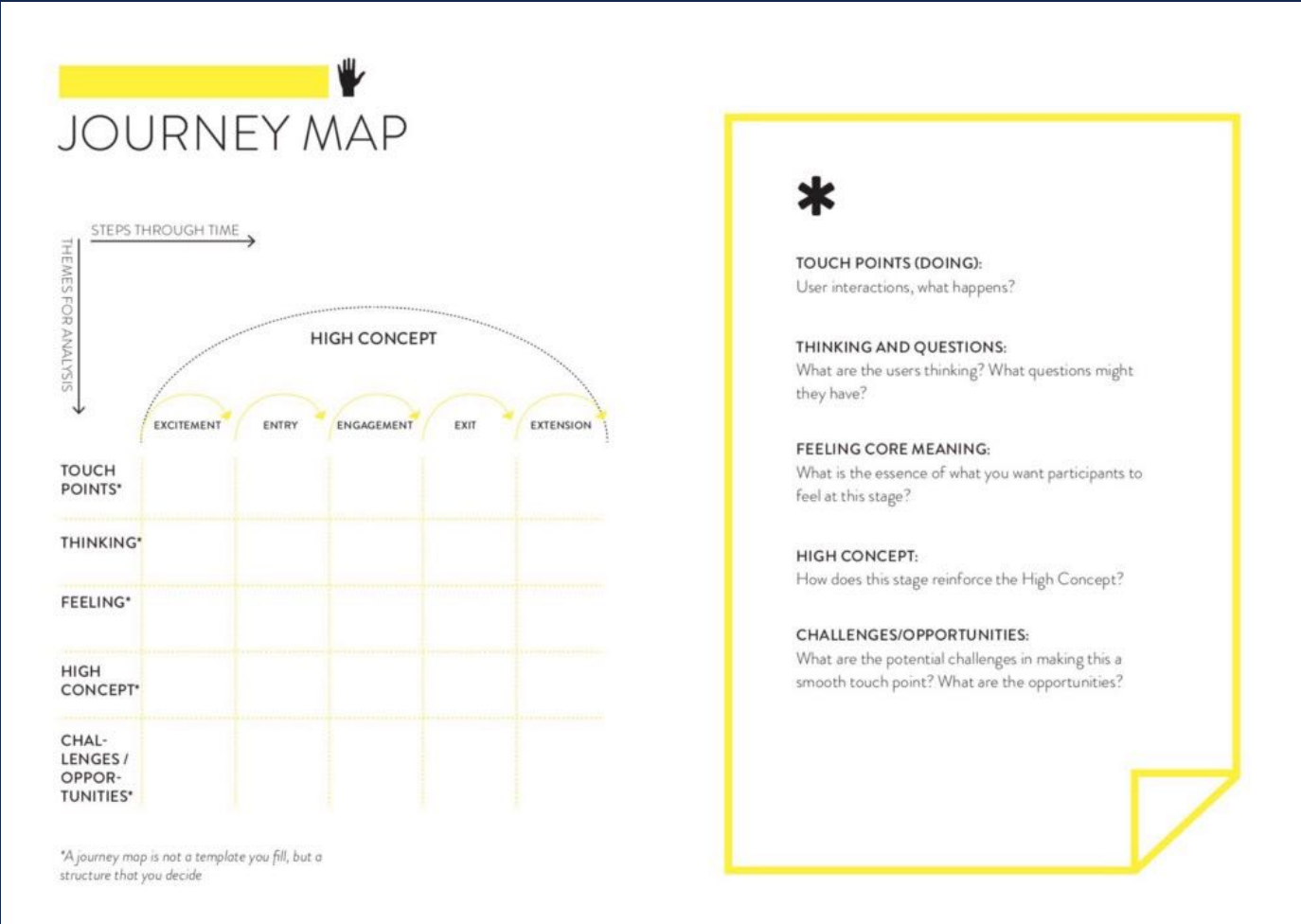
Personas Aren't Enough

- While a persona can improve your user focus, they lacks the details necessary to explain a user and their needs well enough for us to build software that “fits” with their journey.
- The point here is that we need to capture what a user does, how they may encounter a “job to be done”, how our software will assist in solving their issue, and what happens after that interactions concludes.
- Another document should be created that combines personas with activities over time. Usually this is called a *customer journey map*

Customer Journey Model

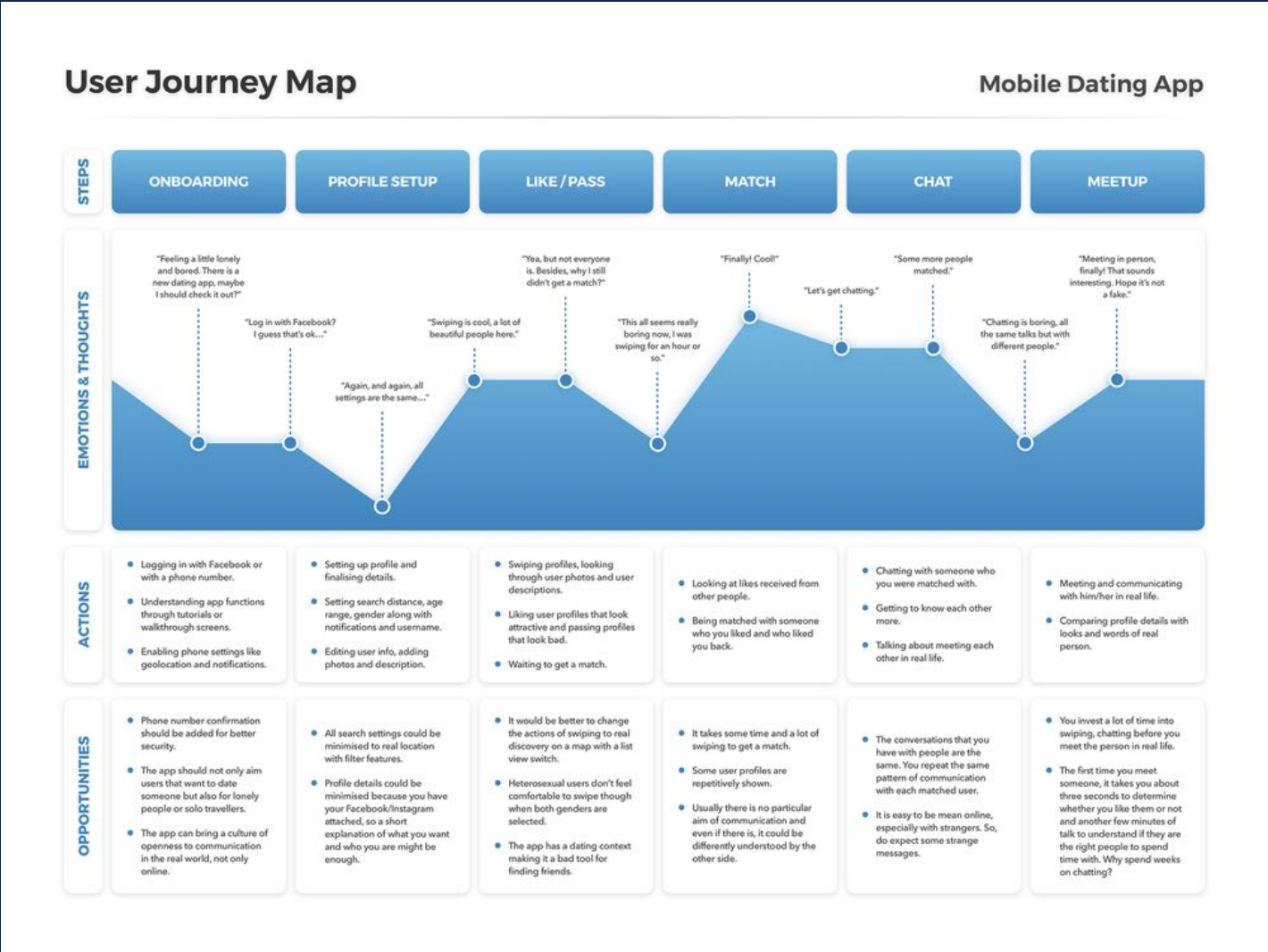
- **5E Model**
 - Entice - How are they attracted or made aware of ____ ?
 - Enter - How do the use begin?
 - Engage - What is the experience like?
 - Exit - How do they complete the experience?
 - Extend - What happens after the experience?

Customer Journey Example



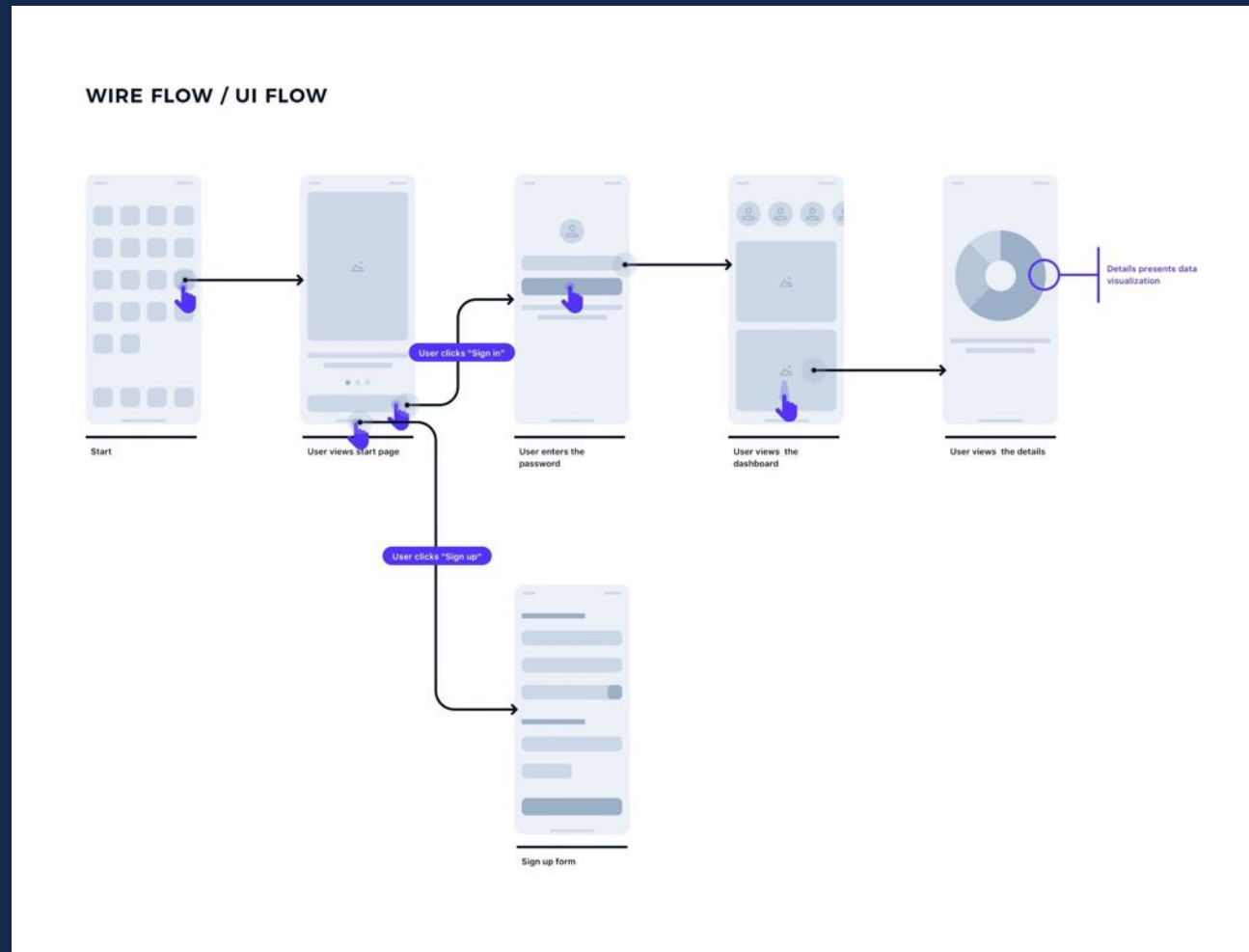
<https://medium.theuxblog.com/the-5e-experience-design-model-7852324d46c>

Customer Journey Map Example



Source: <https://dribbble.com/shots/5950824-Hey-User-Journey-Map>

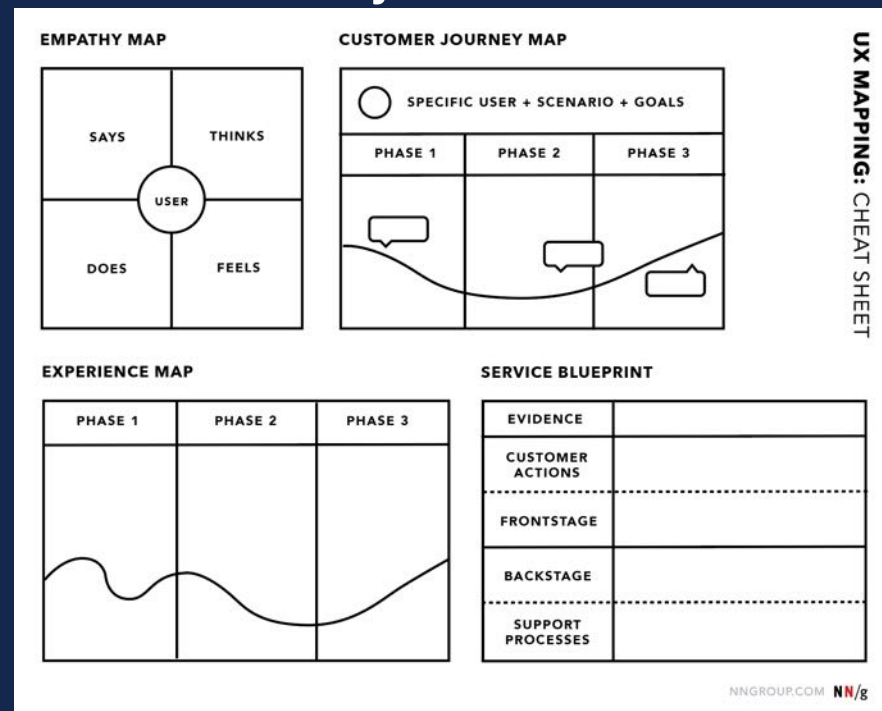
In App Journey: Interface Flow Mapping



<https://uxdesign.cc/a-comprehensive-list-of-ux-design-methods-deliverables-2021-2feb3e70e168>

UX Mapping Diagram Choices

- Actually there are actually many choices to capture how users actually experience the world and your software.
Some possibilities include:



<https://www.nngroup.com/articles/ux-mapping-cheat-sheet/>

Experience Map Example: Many Journeys Across a Customer Relationship

Ecommerce Customer Experience Map

Guiding Principles			
People choose online shopping because it is convenient, easy and comfortable	www.ecommerce.com is only one touch point of an online ecommerce customer experience	Customers are most apprehensive during this process	People appreciate an experience that is prompt, friendly & timely and then talk about it

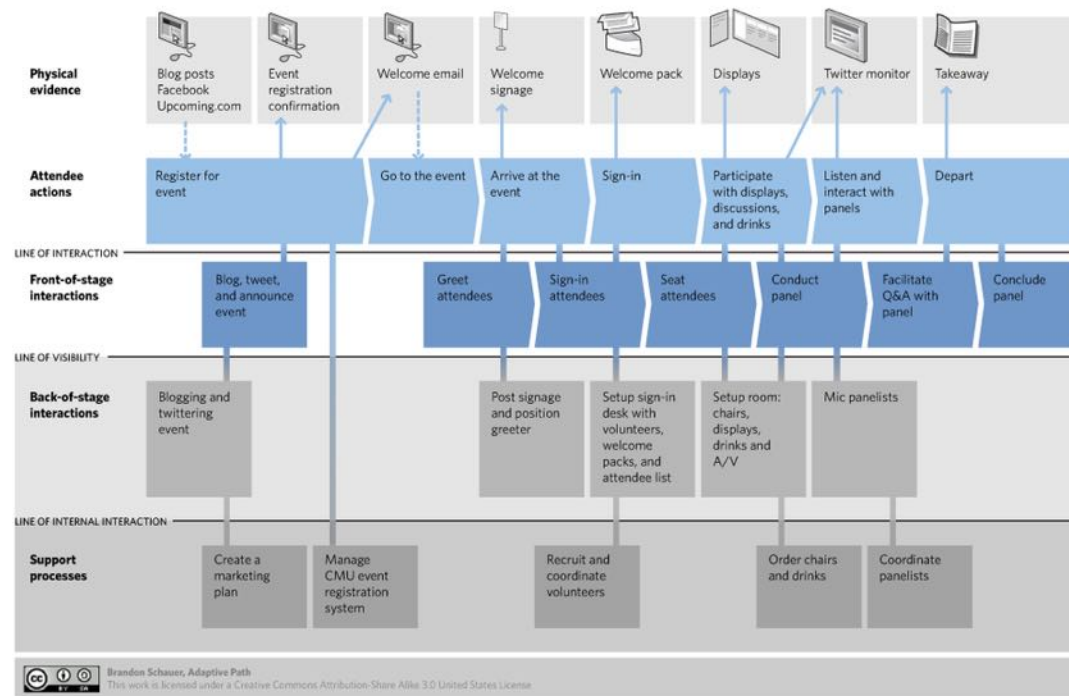
Customer Journey

STAGES	Research & Planning	Product Discovery	Ordering Process	Product Shipping & Delivery	Returns	Refunds
ECOMMERCE CUSTOMER	Research products, prices and offers	Enter product name Click on "BUY NOW" button	Add to shopping Cart Delivery options Payment options Review & confirm	Wait for products to arrive	Satisfied/unexpected issues with product	Share experience Follow-up on refunds
DOING						
THINKING	- What is the easiest way to get this product? - Where should I order it? - Which website is giving the best offer possible? - What are the specifications of the product?	- I want to get the best price, but I'm willing to pay a little more if I can get it fast. - Will they deliver it at my house? - Are there any other offers that I can get with this product?	- Can I trust this website? - Should I have to pay extra shipping charges? - Ecommerce is not answering the phone. How else can I get my question answered?	- Do I have everything I need? - Ecommerce website was easy and friendly, but when an issue came up, I couldn't get help. - What will I do if my product doesn't arrive in time?	- There were some issues with my product. What can we do now? - Did I get the right product? If not, what next? - I want to make more purchases. How do I do that?	- Trying to return product I was not able to use. Not sure if I'll get a refund or not. - My friends may love to order this product! - Next time, I will check the prices and product details more carefully.
FEELING	- I want to buy this product. - Will this product offer everything I need? - Will I be able to afford this pro? - What are friends/blogs/google saying about this product?	- The product looks really awesome on the website. - Specifications of the product seems to be suiting my requirement. - Am I sure this is the product I want to buy?	- Website experience is easy and friendly! - Product is out of stock. - Frustrated to not know sooner that - The product is not delivered in my area. - The payment option, I want to use is not available. - Not sure my products will arrive in time.	- Stressed that - My product has not yet come and Ecommerce won't answer the phone. - My order got cancelled without informing me. - I am not aware about the status of my order.	- I am happy with the product I received. - The product that I received is not upto my expectation. - Interacting with customer service agents was helpful. - I did not get the Freebie promised with the product.	- Excited to share my product experience with my friends. - A bit annoyed to be dealing with refund issues and wasting my time.
EXPERIENCE	Enjoyability Helpfulness of Ecommerce	Enjoyability Helpfulness of Ecommerce	Enjoyability Helpfulness of Ecommerce	Enjoyability Helpfulness of Ecommerce	Enjoyability Helpfulness of Ecommerce	Enjoyability Helpfulness of Ecommerce

Example: Service Blueprint

Service Blueprint for Seeing Tomorrow's Services Panel

find out more: <http://upcoming.yahoo.com/event/1768041>



https://en.wikipedia.org/wiki/Service_blueprint

This shows the “touch points” and also we delicate front and back operations

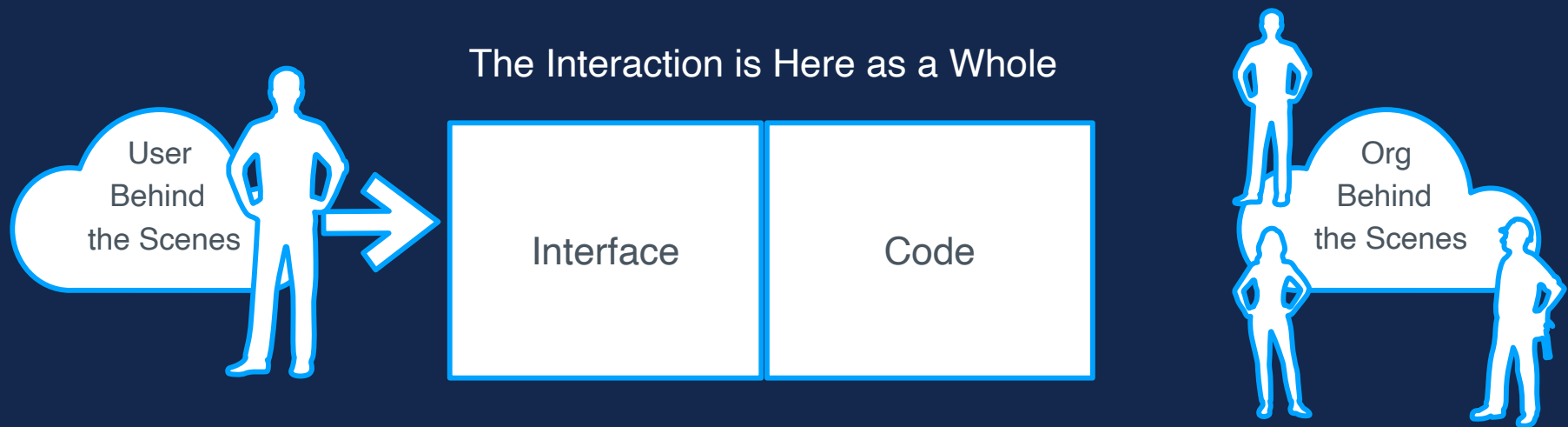
Focus on User Touch Points

- It is important to note all the places that a user touches your system or interacts with your brand as the tone of an offering should be consistent across touch points regardless of medium or location
 - In digital-analog offerings this might be broad
 - In all digital offerings you should differentiate each of the contexts such as email messages, web site, app, etc.

Q: Why must we map touch points?

A: To Make Sure We Consistently Match User Expectations

Also Set the “Stage” for Interaction



A common concept found in restaurants (Front of the House - FOH and Back of the House - BOH), performing arts, etc. is a useful way to think about our code's relationship with users

Always Build Near User “Roads”

- The various steps they take may not be physical always, but as they move from a problem to a solution end users take a **journey**.
- If your product is to succeed you have to be near or on the path of a user's journey.
- Think like a gas station in the middle of nowhere as opposed to on the main highway. Clearly a well positioned station will do well and an absurd one finds it is near no roads and thus dies despite any constructive quality or utility.

Make Sure to Capture Journey Feelings?

- Such an idea sounds right, but how do we do it?
- Visuals and words that tell a story with feelings in it

Chad is stressed, he has a big assignment due tomorrow in a class that is very important for his graduation. Unfortunately all the distractions around him was making focus difficult so he launched PomoTimer. The launch sound was a calming tone as a color sequence cycled subtly as a prompt to work. The software's elegant simplicity was not another distraction for Chad. He thought he would start the slide review first so he pressed the timer button and the work session wants to gain focus so he has

Since we are on the topic of stories...time for the next artifact

Project Artifact: User Stories

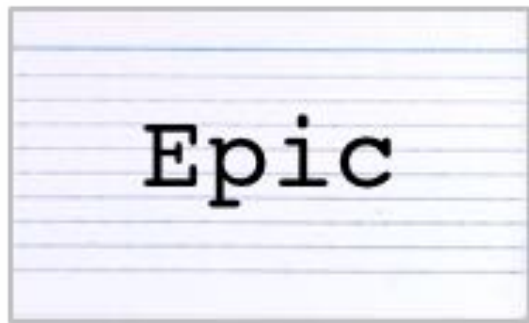
Agile's User Stories

- A formal approach to user centered thinking is a key component of the Agile process and is known as user stories
- Examples
 - “As a home bound remote developer I want a software system to help me keep track of my tasks”
 - “As a home bound remote developer I want a software aid to help me reclaim my focus so I can get more done.”

User Stories

- A user story intends to encourage developers to design tasks to perform and features to add with a user in mind
- The form of a user story is generally thought of as
 1. **As a** [user who is the focus on the story]
 2. **I want** [to accomplish the goal of the story]
 3. **so that** [reason for the story].

Start with Epics End with Stories



Big, coarse-grained,
sketchy

Small, detailed,
specific

User Story Template

- User Story
 - **As a** *<type of user>*
 - **I want to** *<perform some task>*
 - **So that I can** *<achieve some goal>*
- Acceptance Criteria
 - **Given** *<some context>*
 - **When** *<some action is carried out>*
 - **Then** *<a set of observable outcomes should occur>*

User Story Template - in Action

- User Story
 - **As an** *easily distracted employee*
 - **I want to** *make sure all my work gets done for the day*
 - **So that I can** *finish work early and have more free time*
- Acceptance Criteria
 - **Given** *an employee wants to get all their work done that day,*
 - **When** *they open the app*
 - **Then** *show them just what they have left to complete that day*

Another User Story Template

- Story
- Design
 - Approach
 - Concerns
 - Plan (Mockups, Diagrams)
- Implementation
 - Tasks
 - Tests
 - Documentation
- Resources

**Story + Details
= Success**

User Story Criticisms

- Concern: The format is wordy and makes you wade thru filler words to get to the details
- Response(s)
 - The format is predictable and widely used
 - The format forces you not to slip back into features or execution first thinking by making you always write about “users”
 - If user stories annoys you do you think it should be done at all or just not by you? The answer to this question is important to uncover customer centeredness gaps and specialization realities.

Example User Stories

- Zoom Story
 - As a user of teacher under remote conditions, I want to be able to use transitions so that my lectures are more professional and engaging.
- Canvas Story
 - As a student under remote conditions, I want my quiz grades to be delivered quickly with accuracy, so that I am not stressed out about my grades.
- Class Story - Shows Contrast of User Stories
 - As a student of CSE 110, I want clarity of information, so that I can more easily score a good mark in the course.
 - As a Professor of CSE 110, I want ambiguity in some places, so that students begin to take ownership of the software development process.

User Story Criticisms

- Concern: What about the business/owners/developers? They matter too!
- Responses:
 - Actually very good point they must be part of this equation as well, without them we likely don't have a truly realistic system that is economically viable.
 - You can use user stories for these people as well
 - As a developer I want to work with relevant technology like X, so that I am gainfully employed in the future.
 - As the project stake holder I want to minimize my risk of project cost overruns or catastrophic bugs, so that I am gainfully employed in the future.
- Question: Are these two user stories potentially at tension with each other?

User Story Actor Variation Examples

AS A/AN	I WANT TO...	SO THAT...
moderator	create a new game by entering a name and an optional description	I can start inviting estimators
moderator	invite estimators by giving them a url where they can access the game	we can start the game
estimator	join a game by entering my name on the page I received the url for	I can participate
moderator	start a round by entering an item in a single multi-line text field	we can estimate it
estimator	see the item we're estimating	I know what I'm giving an estimate for
estimator	see all items we will try to estimate this session	I have a feel for the sizes of the various items
moderator	see all items we try to estimate this session	I can answer questions about the current story such as "does this include ____"
moderator	select an item to be estimated or re-estimated	the team sees that item and can estimate it

User Story Criticisms

- Concern: How can I actually build software from these little stories?
- Responses
 - You can't actually, but they get the process started.
 - Reality is you need to translate the stories into features of the product and relate things to the system and code.
 - The user story is just a type of requirement that then leads to the specification, designs and execution plan that will lead to code

Expanding the Story

Title:	Priority:	Estimate:
As a <i><type of user></i> I want to <i><perform some task></i> so that I can <i><achieve some goal></i>		
Acceptance criteria Given <i><some context></i> When <i><some action is carried out></i> Then <i><a set of observable outcomes should occur></i>		

Applying Stories to Features and Scenarios

- We can add scenarios to make user stories more translatable to potential features and coding activities

```
Feature: <title>  
  As a <who>  
  I want to <what>  
  Because <why>
```

```
Scenario: <title>  
  Given <state>  
  When <event>  
  Then <expected outcome>  
  And <extend previous  
step>
```

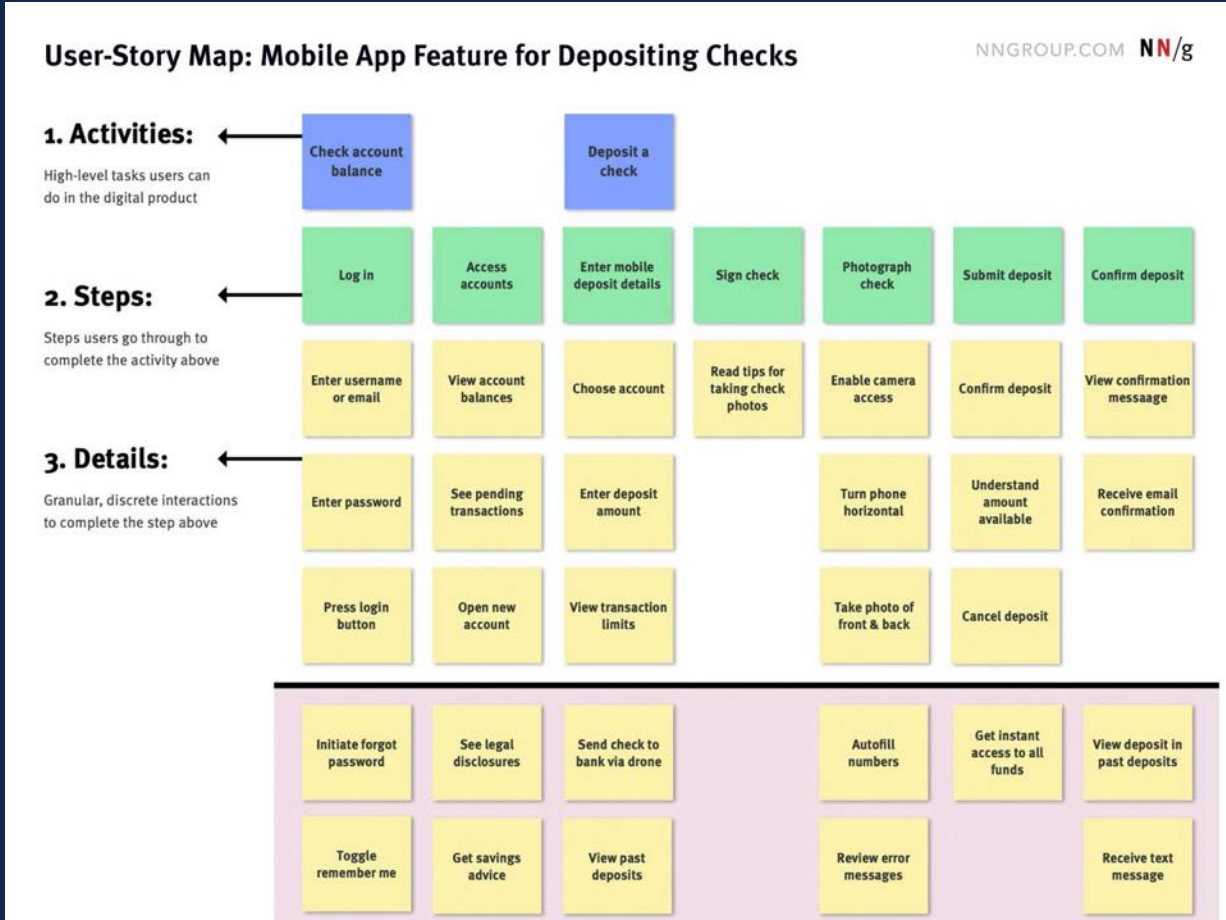
```
Feature: Shopping Cart  
  As a User  
  I want to have a shopping cart  
  Because I need to keep track of  
items for purchase
```

```
Scenario: Add item to shopping cart  
  Given I'm a user  
  When I visit a product page  
  And I click 'Add to cart'  
  Then the item should be added to my  
cart  
  And I should go to my shopping cart
```

User Story and Customer Journey Mapping

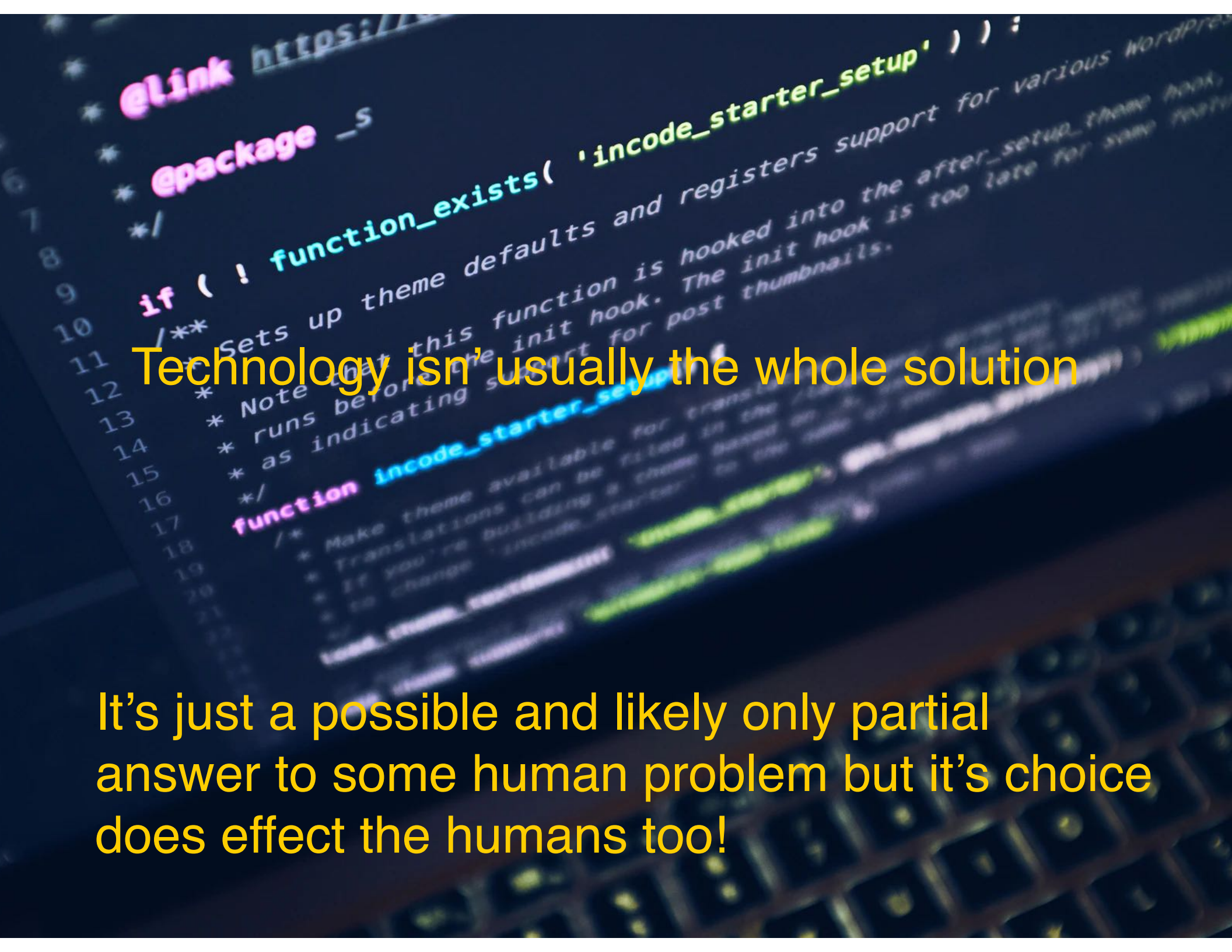
- By themselves individual user stories may not tell a complete story about how a person uses your software.
- You might address this gap by making a diagram called a user story map which is related to customer journey maps
- The idea of the user story map is to show how high level stories (activities) might consist of many steps and have details which support those steps.

User Story Map Example



<https://www.nngroup.com/articles/user-story-mapping/>

Acknowledge Tech Cause and Effect To Help Build a Long Term Project Roadmap



Technology isn't usually the whole solution

It's just a possible and likely only partial answer to some human problem but it's choice does effect the humans too!

Situation



Question: What human problem did this solve?



Problem Solved?

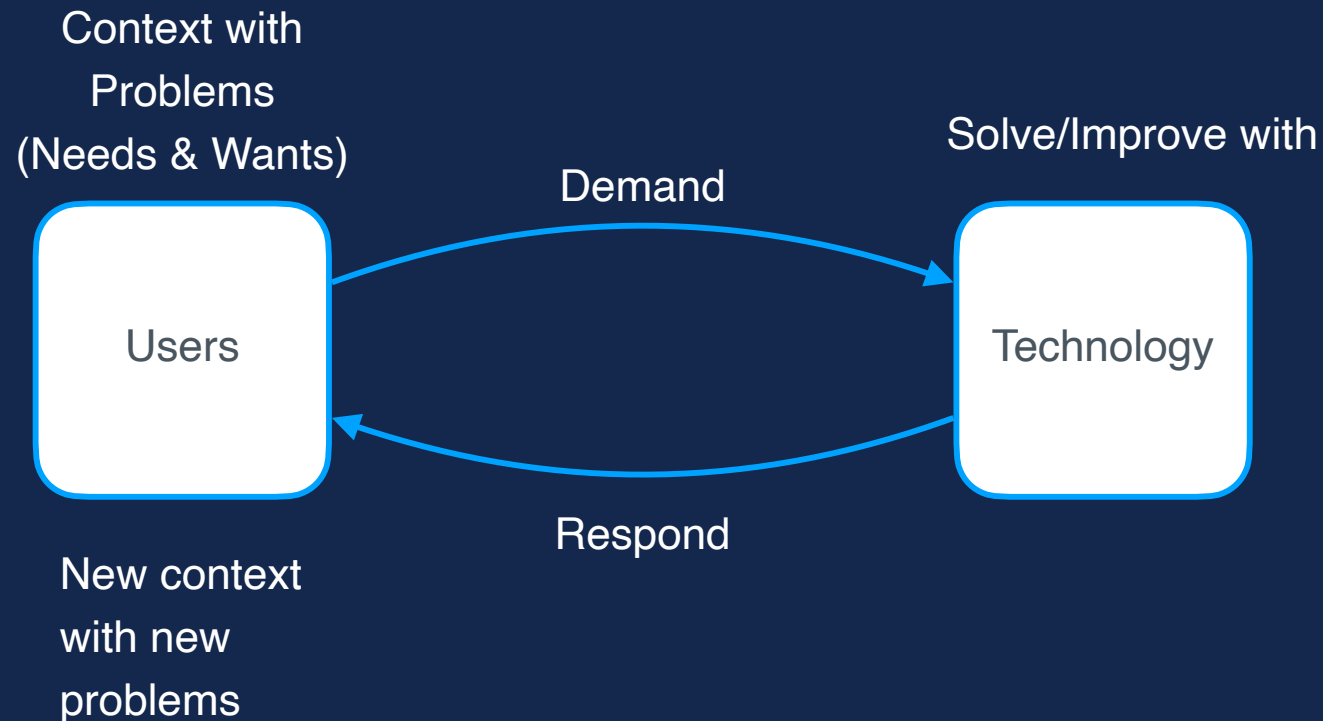


Tech Solution Cause and Effect



Not every effect is positive or negative

Problem Solving Creates New Circumstances



Change of Solution Form Changes Circumstances and Repeat



Think of the User Outcomes



Really Back Off and See Broader Systems



Cause and effect can be complex and include things you don't think of from economics and culture far outside the technical problem.

Design Process Conclusion

- You know you did enough when you either have run out of time (time boxed) or the new ideas aren't coming as fast as they used to.
- In our case it is likely to be a little bit of both, but make sure you don't short change your development by under doing your design effort. In normal projects well beyond software a planning budget of 10%, 15%, even 25% is not that uncommon. Work backwards from time available to see you might want to spend a full week on this!
- The big picture and second and third order effects I just hinted at really aren't out of line given that thought

Summary

- The exploration process you should employ should mirror the double diamond approach to some degree
- Project artifacts should be produced to get alignment within and outside the team on exactly what the problem is, who it is for and what characteristics might the solution take
- The Agile process favors user thinking at all steps and if we do this process right we will be easy to populate our backlog, start designing a slice (sprint) to do, and begin to make our pitch to start the coding part of the job!